

Part IV

Auxiliary Components and Accessories

13 Bearings and Seals

Bearings

The bearings in gas and steam turbines provide support and positioning for the rotating components. Radial support is generally provided by journal, or roller bearings, and axial positioning is provided by thrust bearings. Some engines, mainly aircraft jet engines, use ball or roller bearings for radial support, but nearly all-industrial gas turbines use journal bearings.

A long service life, a high degree of reliability, and economic efficiency are the chief aims when designing bearing arrangements. To reach these criteria, design engineers examine all the influencing factors:

1. Load and speed
2. Lubrication
3. Temperatures
4. Shaft arrangements
5. Life
6. Mounting and dismounting
7. Noise
8. Environmental conditions

Rolling Bearings

The aero-derivative gas turbine design, with its low-supported weight rotors – for example, the LM 5,000 HP rotor weighs 1,230 lbs (558 kg) – incorporates roller bearings throughout. These do not require the large lube oil reservoirs, coolers and pumps, or the pre- and post-lube cycle associated with journal bearing designs. Roller bearings have proven to be extremely rugged and have demonstrated excellent life in industrial service. Most bearings provide reliable service for over 100,000 hours. In practice, it is advisable to replace bearings when exposed during major repairs, estimated at 50,000 hours for gas generators and 100,000 hours for power turbines.

There are many roller bearing types. They are differentiated according to the direction of the main radial loads (radial bearings) or axial loads (thrust bearings), and the type of rolling elements used, balls or rollers. [Figure 13-1](#) shows the different types of bearings. The essential difference between ball bearings and roller bearings are that ball bearings have a lower carrying capacity and higher speeds, while roller bearings have higher load carrying capacity and lower speeds.

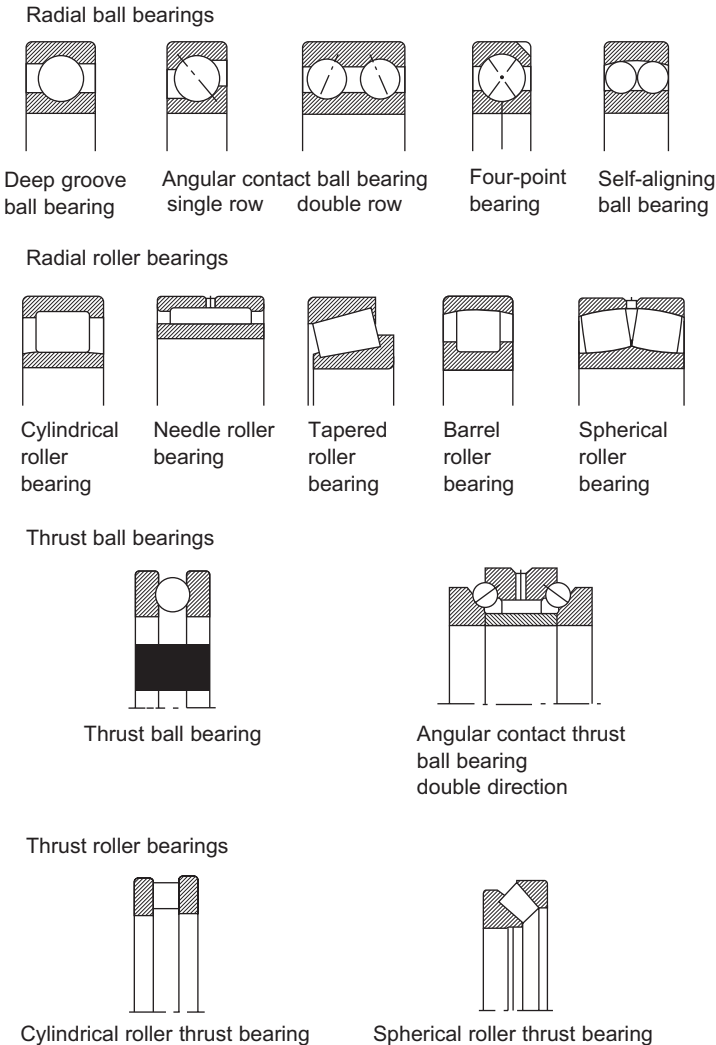


Figure 13-1 Types of rolling bearings (courtesy FAG bearings).

The rolling elements transmit loads from one bearing ring to the other in the direction of the contact lines. The contact angle α is the angle formed by the contact lines and the radial plane of the bearing. α refers to the nominal contact angle, i.e., the contact angle of the load-free bearing as seen in [Figure 13-2](#). Under axial loads, the contact angle of deep groove ball bearings, angular contact ball bearings, etc., increases. Under a combined load it changes from one rolling element to the next. These changing contact angles are taken into account when calculating the pressure distribution within the bearing. Ball bearings and roller bearings with symmetrical rolling elements have identical contact angles at their inner rings and outer rings. In roller bearings with asymmetrical rollers,

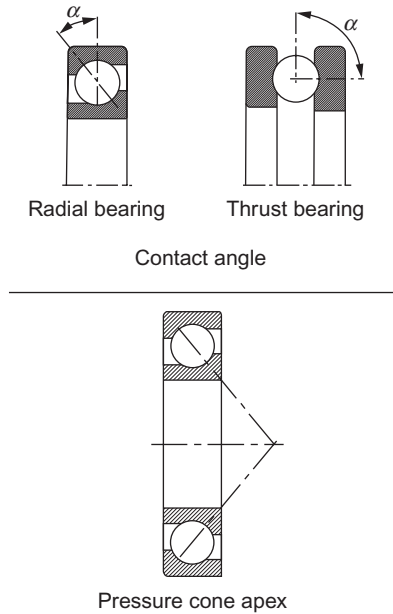
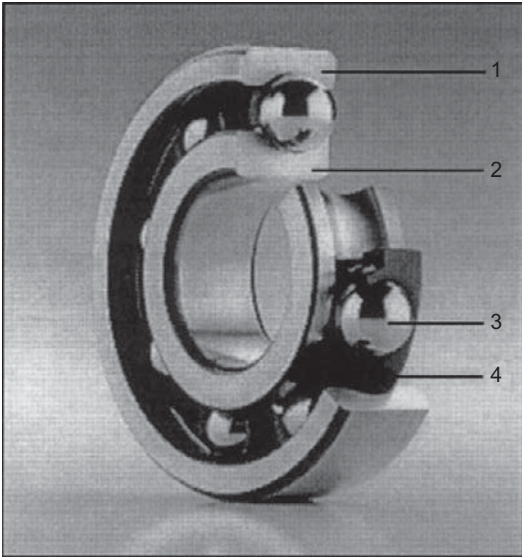


Figure 13-2 Rolling bearing terminology (courtesy FAG bearings).

the contact angles at the inner rings and outer rings are not identical. The equilibrium of forces in these bearings is maintained by a force component, which is directed towards the lip. The pressure cone apex is that point on the bearing axis where the contact lines of an angular contact bearing (i.e., an angular contact ball bearing, a tapered roller bearing, or a spherical roller thrust bearing) intersect. The contact lines are the generatrices of the pressure cone apex. In angular contact bearings, the external forces act, not at the bearing center, but at the pressure cone apex.

Rolling bearings generally consist of bearing rings, inner ring and outer ring, rolling elements that roll on the raceways of the rings, and a cage that surrounds the rolling elements as seen in [Figure 13-3](#). The rolling elements are classified according to their shapes, into: balls, cylindrical rollers, needle rollers, tapered rollers, and barrel rollers as shown in [Figure 13-4](#).

The rolling elements' function is to transmit the force acting on the bearing from one ring to the other. For a high load carrying capacity it is important that as many rolling elements as possible, which are as large as possible, are accommodated between the bearing rings. Their number and size depend on the cross section of the bearing. It is just as important for load ability that the rolling elements within the bearing are of identical size. Therefore, they are sorted according to grades. The tolerance of one grade is very slight. The generatrices of cylindrical rollers and tapered rollers have a logarithmic profile. The center part of the generatrix of a needle roller is straight, and the ends are slightly crowned, this profile prevents edge stressing when under load.



(1) Outer ring (2) Inner ring (3) Rolling element and (4) Cage

Figure 13-3 Roller bearing showing the various components (courtesy FAG Bearings).

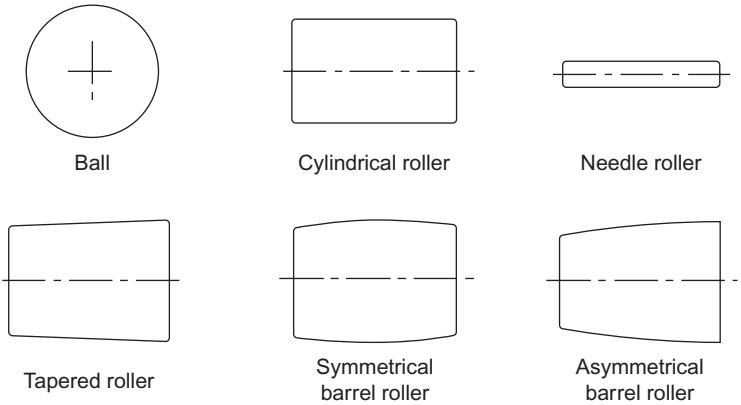


Figure 13-4 Various types of rollers used in a rolling bearing (courtesy FAG bearings).

The bearing rings comprise an inner ring and an outer ring to guide the rolling elements in the direction of rotation. Raceway grooves, lips, and inclined running areas guide the rollers and transmit axial loads in a transverse direction as seen in [Figure 13-5](#). Cylindrical roller bearings and needle roller bearings, which need to accommodate shaft expansions have lips only on one bearing ring and are commonly known as floating bearings.

The functions of a cage are to keep the rolling elements apart so that they do not rub against each other, to keep the rolling elements evenly spaced for uniform load

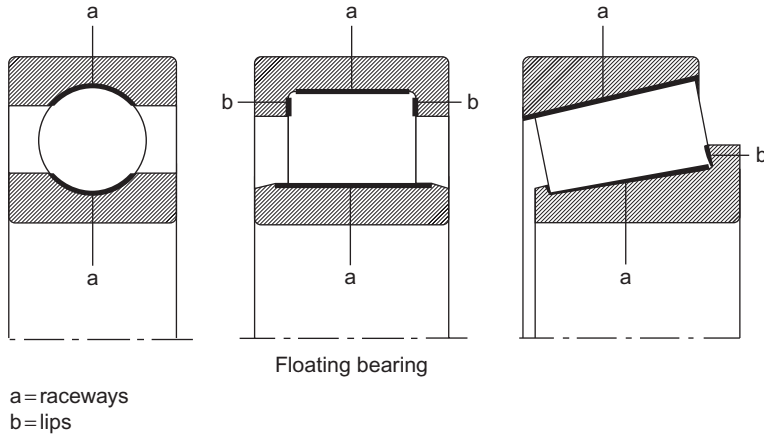


Figure 13-5 Raceway grooves and lips for typical roller bearings (courtesy FAG bearings).

distribution, to prevent rolling elements from falling out of separable bearings and bearings that are swiveled out, and to guide the rolling elements in the unloaded zone of the bearing. The transmission of forces is not one of the cage's functions.

Cages are classified into pressed cages, machined cages, and molded cages. Pressed cages are usually made of steel, but sometimes of brass too. They are lighter than machined metal cages. Since a pressed cage barely closes the gap between the inner ring and outer ring, lubricant can easily penetrate into the bearing.

Machined cages of metal and textile laminated phenolic resin are made from tubes of steel, light metal or textile laminated phenolic resin, or cast brass rings. To obtain the required strength, large, heavily loaded bearings are fitted with machined cages. Machined cages are also used where lip guidance of the cage is required. Lip-guided cages for high-speed bearings are, in many cases, made of light materials such as light metal or textile laminated phenolic resin to keep the forces of gravity low.

Molded cages using injection molding techniques can realize designs with an especially high-load carrying capacity. The elasticity and low weight of the cages are of advantage where shock-type bearing loads, great accelerations and decelerations as well as tilting of the bearing rings relative to each other have to be accommodated. Polyamide cages feature very good sliding and dry running properties.

There are a number of special rolling bearing designs and some series of cylindrical roller bearings without cages. By omitting the cage the bearing can accommodate more rolling elements. This yields an increased load rating, but, due to the increased friction, the bearing is suitable for lower speeds only.

Load ratings

The load rating of a bearing reflects its load carrying capacity and is an important factor in the dimensioning of rolling bearings. It is determined by the number and size of the rolling elements, the curvature ratio, the contact angle, and the pitch circle

diameter of the bearing. Due to the larger contact area between rollers and raceways the load ratings of roller bearings are higher than those of ball bearings.

The load rating of a radial bearing is defined by radial loads whereas that of a thrust bearing is defined by axial loads. Every rolling bearing has a dynamic load rating and a static load rating. The terms “dynamic” and “static” refer to the movement of the bearing but not to the type of load.

In all rolling bearings with a curved raceway profile, the radius of the raceway is slightly larger than that of the rolling elements. This curvature difference in the axial plane is defined by the curvature ratio x . The curvature ratio is the curvature difference between the rolling element radius and the slightly larger groove radius:

$$\text{radius curvature ratio } x = (\text{groove radius} - \text{rolling element}) / \text{rolling element radius}$$

Thrust ball bearings are used where purely axial loads have to be accommodated. The single direction (= single row) design is designed for loads from one direction, the double direction one (= double row) for reversing loads. Besides the design with flat washers, designs with spherical housing washers and seating washers are also available that can compensate for misalignment.

Spherical roller thrust bearings can accommodate high axial loads. They are suitable for relatively high speeds. The raceways, which are inclined toward the bearing axis, allow the bearings to accommodate radial loads as well. The radial load must not exceed 55% of the axial load.

The bearings have asymmetrical barrel rollers and compensate for misalignment. As a rule, spherical roller thrust bearings have to be lubricated with oil.

Wear

The life of rolling bearings can be terminated, apart from fatigue, as a result of wear. The clearance of a worn bearing gets too large.

One frequent cause of wear is that foreign particles penetrate into a bearing due to insufficient sealing and have an abrasive effect. Wear is also caused by starved lubrication and when the lubricant is used up.

Therefore, wear can be considerably reduced by providing good lubrication conditions (viscosity ratio $x > 2$ if possible) and a good degree of cleanliness in the rolling bearing. Where $x \leq 0.4$ wear will dominate in the bearing if it is not prevented by suitable additives (EP additives).

The kinematically permissible speed may be higher or lower than the thermal reference speed. The basis of the thermal reference speed is for cases where the operating conditions (load, oil viscosity, or permissible temperature) deviate from the reference conditions. Decisive criteria for the kinematically permissible speed are the strength limit of the bearing parts and the permissible sliding velocity of rubbing seals. Kinematically permissible speeds, which are higher than the thermal reference speeds, can be reached, for example, with specially designed lubrication, bearing clearance

adapted to the operating conditions, and accurate machining of the bearing seats with special regard to heat dissipation.

The thermal reference speed is a new index of the speed suitability of rolling bearings. It is defined as the speed at which the reference temperature of 160 °F (70 °C) is established. For high temperature rolling bearings, the steel used for bearing rings and rolling elements is generally heat-treated so that it can be used at operating temperatures of up to 300 °F (150 °C). At higher temperatures, dimensional changes and hardness reductions result. Therefore, operating temperatures over 300 °F (150 °C) require special heat treatment.

Journal Bearings

The heavy frame type gas turbines use journal bearings. Journal bearings may be either full round or split; the lining may be heavy as used in large-size bearings for heavy machinery, or thin, as used in precision insert-type bearings in internal combustion engines. Most sleeve bearings are of the split type for convenience in servicing and replacement. Often in split bearings, where the load is entirely downward, the top half of the bearing acts only as a cover to protect the bearing and to hold the oil fittings. [Figure 13-6](#) shows a number of different types of journal bearings. A description of a few of the pertinent types of journal bearings is given here:

1. *Plain journal.* Bearing is bored with equal amounts of clearance (on the order of $1\frac{1}{2}$ to two thousands of an inch per inch of journal diameter) between the journal and bearing.
2. *Circumferential grooved bearing.* Normally has the oil groove at half the bearing length. This configuration provides better cooling, but reduces load capacity by dividing the bearing into two parts.
3. *Cylindrical bore bearings.* Another common bearing type used in turbines. It has a split construction with two axial oil-feed grooves at the split.
4. *Pressure or pressure dam.* Used in many places where bearing stability is required, this bearing is a plain journal bearing with a pressure pocket cut in the unloaded half. This pocket is approximately $1/32$ of an inch (.8 mm) deep with a width 50% of the bearing length. This groove or channel covers an arc of 135° and terminates abruptly in a sharp-edge dam. The direction of rotation is such that the oil is pumped down the channel toward the sharp edge. Pressure dam bearings are for one direction of rotation. They can be used in conjunction with cylindrical bore bearings as shown in [Figure 13-6](#).
5. *Lemon bore or elliptical.* This bearing is bored with shims at the split line, which are removed before installation. The resulting bore shape approximates an ellipse with the major axis clearance approximately twice the minor axis clearance. Elliptical bearings are for both directions of rotation.
6. *Three-lobe bearing.* The three-lobe bearing is not commonly used in turbomachines. It has a moderate load-carrying capacity and can be operated in both directions.
7. *Offset halves.* In principle, this bearing acts very similar to a pressure dam bearing. Its load-carrying capacity is good. It is restricted to one direction of rotation.
8. *Tilting-pad bearings.* This bearing is the most common bearing type in today's machines. It consists of several bearing pads posed around the circumference of the shaft. Each pad is able to tilt to assume the most effective working position. Its most important feature is

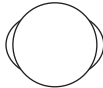
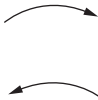
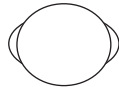
Bearing type	Load capacity	Suitable direction of rotation	Resistance to half-speed whirl	Stiffness and damping
Cylindrical bore 	Good		Worst Increasing Best	Moderate
Cylindrical bore with dammed groove 	Good			Moderate
Lemon bore 	Good			Moderate
Three lobe 	Moderate			Good
Offset halves 	Good			Excellent
Tilting pad 	Moderate			Good

Figure 13-6 Comparison of general bearing types.

self-alignment when spherical pivots are used. This bearing also offers the greatest increase in fatigue life because of the following advantages:

- a. Self-aligning for optimum alignment and minimum limit.
- b. Thermal conductivity backing material to dissipate heat developed in oil film.

- c. A thin babbitt layer can be centrifugally cast with a uniform thickness of about 0.005 of an inch (0.127 mm). Thick babbitts greatly reduce bearing life. Babbitt thickness in the neighborhood of .01 in. (.254 mm) reduce the bearing life by more than half.
- d. Oil film thickness is critical in bearing stiffness calculations. In a tilting-pad bearing, one can change this thickness in a number of ways: (1) changing the number of pads; (2) directing the load on or in between the pads; (3) and changing axial length of pad.

The previous list contains some of the most common types of journal bearings. They are listed in the order of growing stability. All of the bearings designed for increased stability are obtained at higher manufacturing costs and reduced efficiency. The antiwhirl bearings all impose a parasitic load on the journal, which causes higher power losses to the bearings, and in turn, requires higher oil flow to cool the bearing. Many factors enter into the selection of the proper design for bearings. Some of the factors that affect bearing design follow:

1. Shaft speed range.
2. Maximum shaft misalignment that can be tolerated.
3. Critical speed analysis and the influence of bearing stiffness on this analysis.
4. Loading of the compressor impellers.
5. Oil temperatures and viscosity.
6. Foundation stiffness.
7. Axial movement that can be tolerated.
8. Type of lubrication system and its contamination.
9. Maximum vibration levels that can be tolerated.

Bearing Design Principles

The journal bearing is a fluid-film bearing. This description means that a full film of fluid completely separates the stationary bushing from the rotating journal – the two components that make up the bearing system. This separation is achieved by pressurizing the fluid in the clearance space to the extent that the fluid forces a balance in the bearing load. This balance requires the fluid to be continuously introduced into and pressurized in the film space. [Figure 13-7](#) shows the four modes of lubrication in a fluid-film bearing. The hydrodynamic mode bearing is the most common bearing type used and is also often called the “self-acting” bearing.

As can be seen in [Figure 13-7 \(a\)](#), the pressure is self-induced by the relative motion between the two bearing member surfaces. The film is wedge-shaped in this type of lubrication mode. [Figure 13-7 \(b\)](#) shows the hydrostatic mode of lubrication. In this type of a bearing, the lubricant is pressurized externally and then introduced in the bearing. [Figure 13-7 \(c\)](#) shows the squeeze-film lubrication mode. This type of a bearing derives its load-carrying capacity and separation from the fact that a viscous fluid cannot be instantaneously squeezed out between two surfaces that are approaching each other. [Figure 13-7 \(d\)](#) shows a hybrid-type bearing that combines the previous modes. The most common hybrid type combines the hydrodynamic and hydrostatic modes.

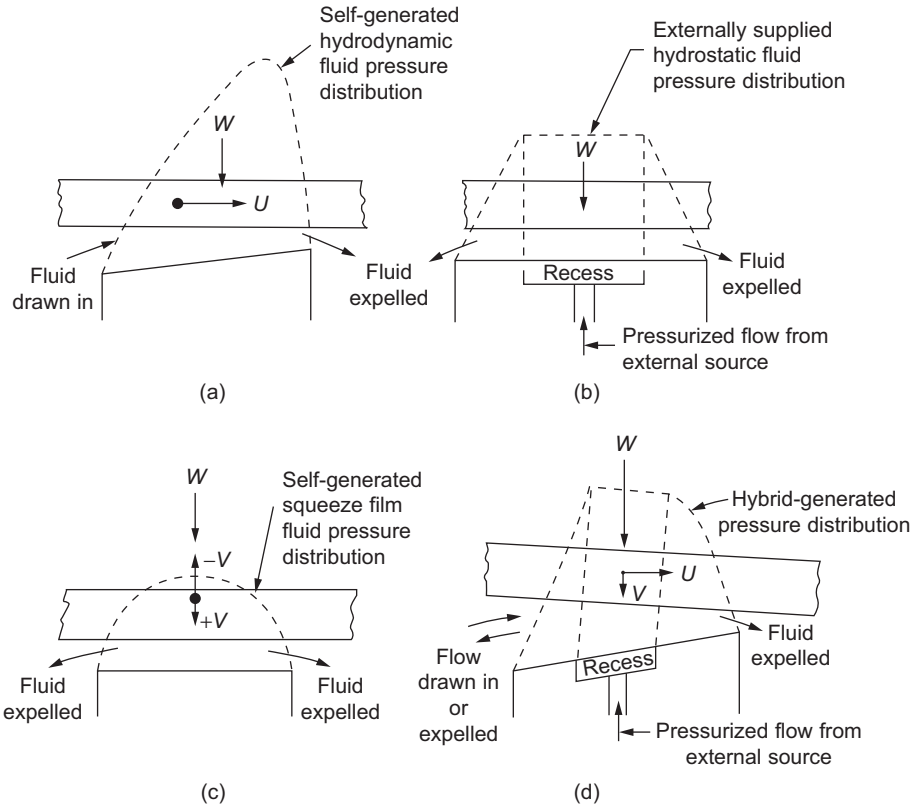


Figure 13-7 Modes of fluid-film lubrication: (a) hydrodynamic, (b) hydrostatic, (c) squeeze film, (d) hybrid.

A further investigation of the hydrodynamic mode is warranted, since it is the most common type of lubrication mode employed. This type of lubrication depends on the bearing member velocity as well as the existence of a wedge-shaped configuration. The journal bearing forms a natural wedge as seen in Figure 13-8, which is inherent in its design. Figure 13-3 also shows the pressure distribution in the bearing. Fluid-film thickness depends on the mode, lubrication, and application and varies from .0001 to .01 inches (.00254–.254 mm). For hydrostatic oil-lubricated bearings, the film thickness is .008 of an inch (.203 mm). In the special case of oil-squeeze film bearings where the capacity must be provided to take extremely high-revising loads with no bearing harm, the oil-film thickness could be below .0001 of an inch (.00254 mm). Since the film thickness is so very important, an understanding of the surface is also vital.

All surfaces, regardless of their finish, are made up of peaks and valleys, and in general, the average asperity height may be 5–10 times the RMS surface finish reading. When the surface is abraded, an oxide film will form almost immediately.

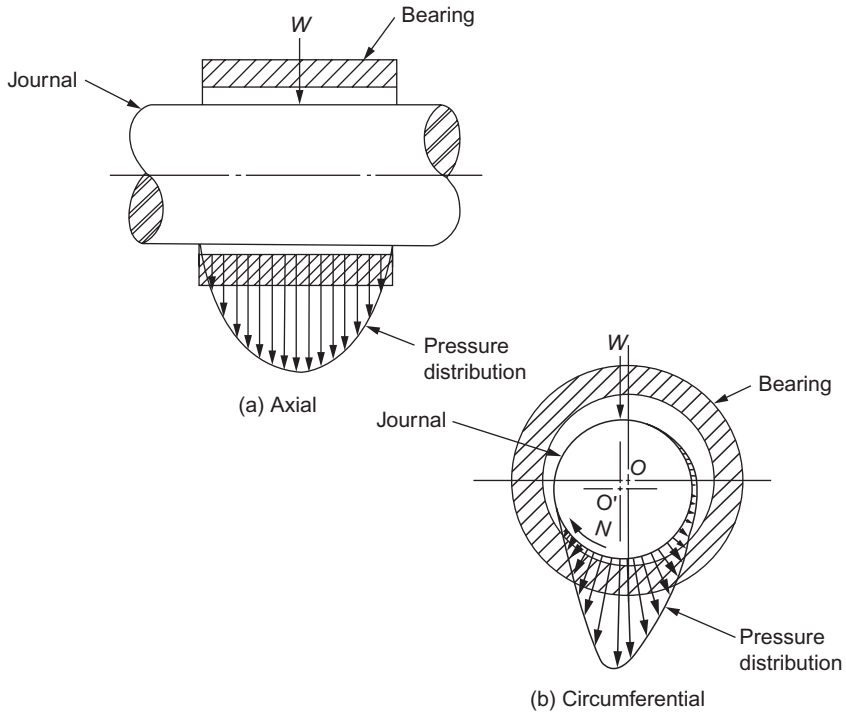


Figure 13-8 Pressure distribution in a full journal bearing.

Figure 13-9 (a) shows the relative separation of the full-film, mixed-film, and boundary. If a full-film exists, the bearing life is almost infinite. The limitation in the case of full-film is due to lubricant breakdown, shock load, bearing surface erosion, and fretting of bearing components. Figures 13-9 (b) and 13-9 (c) are cross sections showing the various contamination types. Oil additives are contaminants that form beneficial surface films.

The bearing health can be best described by plotting a ZN/P versus coefficient of friction curve. Figure 13-10 shows such a curve where Z is the lubricant viscosity in centipoise, N the rpm of the journal, and P is the projected area unit loading.

As the bearing speed is increased for a given lubricant and loading, the friction is at its lowest when full-film is achieved, after which the increase is due to the increasing lubricant shear force.

The bearing fluid film acts like a spring that is nonlinear. Figure 13-11 shows a curve of bearing load versus film thickness and eccentricity ratio. The bearing stiffness can then be obtained at any load value by drawing a line tangent to the curve at the load point. The film stiffness can then be used in determining the critical speed of the rotor.

With higher speeds and unusual fluid lubricants, turbulence in the fluid film is no longer rare. Normally, thin film is thought of as being laminar, but with high speeds,

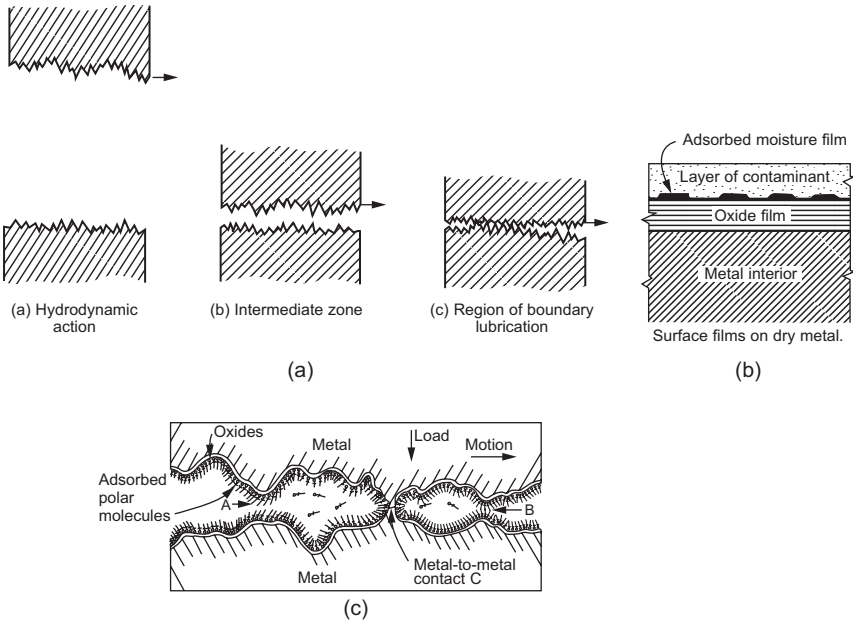


Figure 13-9 Enlarged views of bearing surfaces.

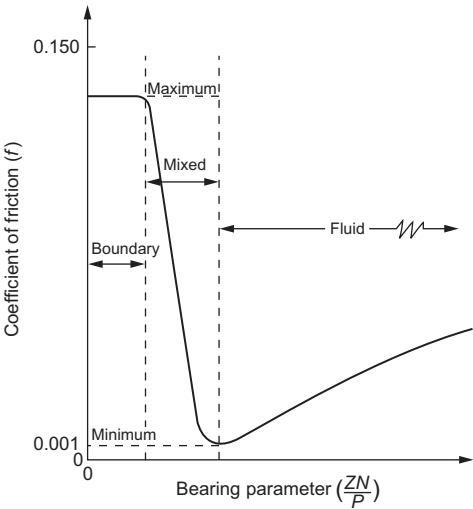


Figure 13-10 Classic ZN/P curve.

low viscosity, and sometimes high-density fluids, the lubricant can be turbulent in the film space. This turbulence manifests itself as an abnormal increase in power loss. As compared to laminar-flow conditions, a Reynolds number, even in the transition region, can double the power and, deep in the turbulent region, can increase the power tenfold. Although this phenomenon, because of its random nature, is difficult to

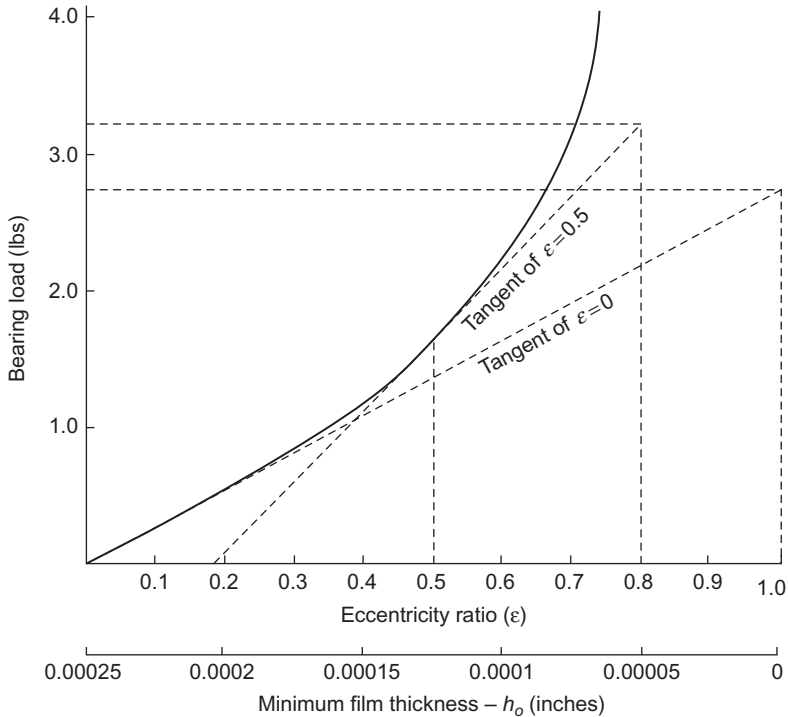


Figure 13-11 Journal bearing load capacity versus minimum film thickness and eccentricity ratio.

analyze, there is an unusual amount of theoretical work that has been done and some experimental work that is available. Just as a guide, one can assume that the transition point will occur at a Reynolds number of about 800. As to film thickness, there is evidence indicating that under turbulent conditions it is actually greater than calculated, based on laminar-flow theory.

Tilting-Pad Journal Bearings

Normally, the tilting-pad journal bearing is considered when shaft loads are light because of its inherent ability to resist oil whirl vibration. However, this bearing, when properly designed, has a very high load-carrying capacity. It has the ability to tilt to accommodate the forces being developed in the hydrodynamic oil film, and therefore operates with an optimum oil-film thickness for the given load and speed. This ability to operate over a large range of loads is especially useful in high-speed gear reductions with various combinations of input and output shafts.

Another important advantage of the tilting-pad journal bearing is its ability to accommodate shaft misalignment. Because of its relatively short length-to-diameter ratio, it can accommodate minor misalignment quite easily.

As shown earlier, bearing stiffness varies with the oil-film thickness so that the critical speed is directly influenced to a certain degree by oil-film thickness. Again, in the area of critical speeds, the tilting-pad journal bearing has the greatest degree of design flexibility. There are sophisticated computer programs that show the influence of various load and design factors on the stiffness of tilting-pad journal bearings. The following variations are possible in the design of tilting-pad bearings:

1. The number of pads can be varied from three to any practical number.
2. The load can be placed either directly on a pad or to occur between pads.
3. The unit loading on the pad can be varied by either adjusting the arc length or the axial length of the bearing pad.
4. A parasitic pre-load can be designed into the bearing by varying the circular curvature of the pad with respect to the curvature of the shaft.
5. An optimum support point can be selected to obtain a maximum oil-film thickness.

On a high-speed rotor system, it is necessary to use tilting-pad bearings because of the dynamic stability of these bearings. A high-speed rotor system operates at speeds above the first critical speed of the system. It should be understood that a rotor system includes the rotor, the bearings, the bearing support system, seals, couplings, and other items attached to the rotor. The system's natural frequency is therefore dependent on the stiffness and damping effect of these components.

Commercial multipurpose tilting-pad bearings are usually designed for multidirectional rotation so that the pivot point is at pad midpoint. However, the design criteria generally applied for producing maximum stability and load-carrying capacity locates the pivot at two-thirds of the pad arc in the direction of rotation.

Bearing pre-load is another important design criterion for tilting-pad bearings. Bearing pre-load is bearing assembly clearance divided by machined clearance:

$$\text{Pre-load ratio} = C'/C = \frac{\text{Concentric pivot film thickness}}{\text{Machined clearance}}$$

A pre-load of 0.5–1.0 provides for stable operation because a converging wedge is produced between the bearing journal and the bearing pads.

The variable C' is an installed clearance and is dependent upon the radial pivot position. The variable C is the machine clearance and is fixed for a given bearing. Figure 13-7 shows two pads of a five-pad tilting-pad bearing where the pads have been installed such that the pre-load ratio is less than one, and Pad 2 has a pre-load ratio of 1.0. The solid line in Figure 13-7 represents the position of the journal in the concentric position. The dashed line represents the journal in a position with a load applied to the bottom pads.

From Figure 13-12, Pad 1 is operating with a good converging wedge, while Pad 2 is operating with a completely diverging film, thus indicating that the pad is completely unloaded. Therefore, bearings with pre-load ratios of 1.0 or greater will be operating with some of their pads completely unloaded, thus reducing the overall stiffness of the bearing and decreasing its stability, since the upper pads do not aid in resisting cross-coupling influences.

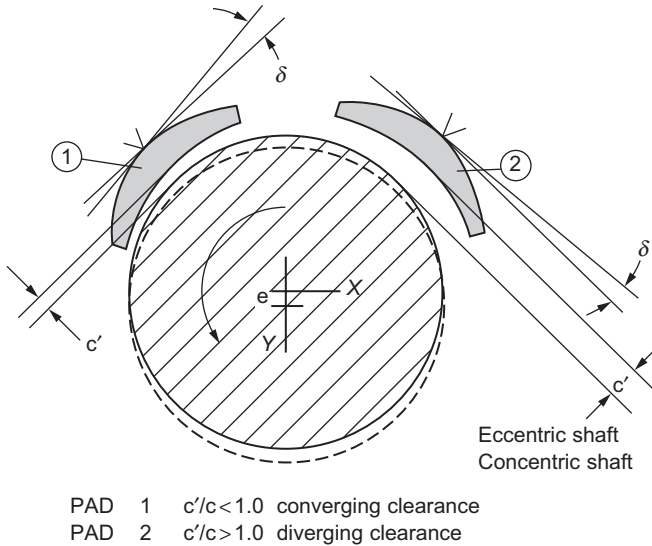


Figure 13-12 Tilting-pad bearing pre-load.

Unloaded pads are also subject to flutter, which leads to a phenomenon known as “leading-edge lock-up.” Leading-edge lock-up causes the pad to be forced against the shaft, and it is then maintained in that position by the frictional interaction of the shaft and the pad. Therefore, it is of prime importance that the bearings be designed with pre-load, especially for low-viscosity lubricants. In many cases, manufacturing reasons and the ability to have two-way rotation cause many bearings to be produced without pre-load.

Bearing designs are also affected by the transition of the film from a laminar to a turbulent region. The transition speed (N_t) can be computed using the following relationship:

$$N_t = 1.57 \times 10^3 \frac{v}{\sqrt{DC^3}}$$

where:

v = viscosity of the fluid

D = diameter (inches)

C = diametrical clearance (inches)

Turbulence creates more power absorption, thus increasing oil temperature that can lead to severe erosion and fretting problems in bearings. It is desirable to keep the oil discharge temperature below 170 °F (77 °C), but with high-speed bearings, this ideal may not be possible. In those cases, it is better to monitor the temperature difference between the oil entering and leaving as shown in [Figure 13-13](#).

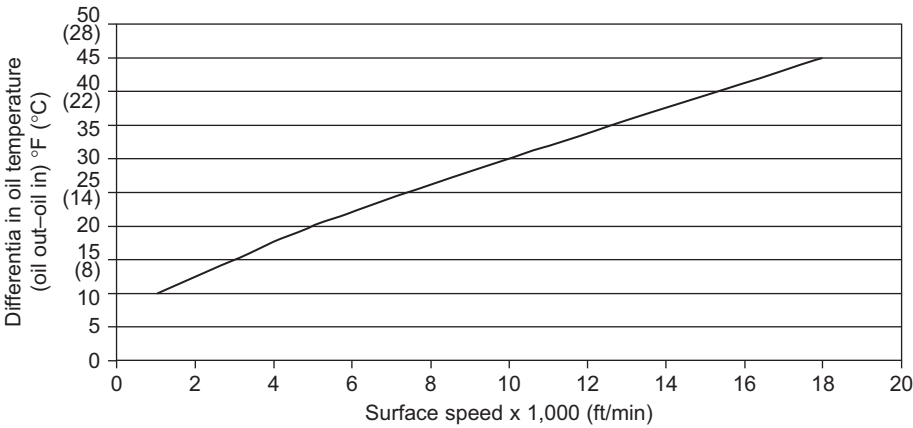


Figure 13-13 Oil discharge characteristics.

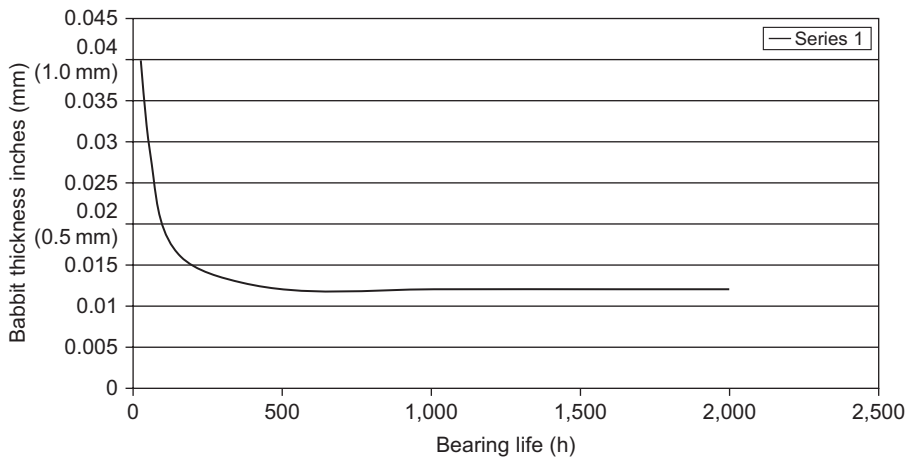


Figure 13-14 Babbitt fatigue characteristics.

Bearing Materials

In all the time since Issaac Babbitt patented his special alloy in 1839, nothing has been developed that encompasses all of its excellent properties as an oil-lubricated bearing surface material. Babbitts have excellent compatibility and non-scoring characteristics and are outstanding in embedding dirt and conforming to geometric errors in machine construction and operation. They are, however, relatively weak in fatigue strength, especially at elevated temperatures and when the babbitt is more than about 0.015 of an inch (.381 mm) thick as seen in [Figure 13-14](#). In general, the selection of a bearing material is always a compromise, and no single composition can include all desirable properties. Babbitts can tolerate momentary rupture of the oil film, and

may well minimize shaft or runner damage in the event of a complete failure. Tin babbits are more desirable than the lead-based materials, since they have better corrosion resistance, less tendency to pickup on the shaft or runner, and are easier to bond to a steel shell.

Application practices suggest a maximum design temperature of about 300 °F (149 °C) for babbitt, and designers will set a limit of about 50 °F (28 °C) less. As temperatures increase, there is a tendency for the metal to creep under the softening influence of the rising temperature. Creep can occur with generous film thickness and can be observed as ripples on the bearing surface where flow took place. With tin babbits, observation has shown that creep temperature ranges from 375 °F (190 °C) for bearing loads below 200 psi (13.79 Bar) to about 260–270 °F (127–132 °C) for steady loads of 1,000 psi (69 Bar). This range may be improved by using very thin layers of babbitt such as in automotive bearings.

Bearing and Shaft Instabilities

One of the most serious forms of instability encountered in journal bearing operation is known as “half-frequency whirl.” It is caused by self-excited vibration and characterized by the shaft center orbiting around the bearing center at a frequency of approximately half of the shaft rotational speed as shown in [Figure 13-15](#).

As the speed is increased, the shaft system may be stable until the “whirl” threshold is reached. When the threshold speed is reached, the bearing becomes unstable, and further increase in speed produces more violent instability until eventual seizure results. Unlike an ordinary critical speed, the shaft cannot “pass through,” and the instability frequency will increase and follow that half ratio as the shaft speed is increased. This type of instability is associated primarily with high-speed, lightly loaded bearings. At present, this form of instability is well understood, can be theoretically predicted with accuracy, and avoided by altering the bearing design.

It should be noted that the tilting-pad journal bearing is almost completely free from this form of instability. However, under certain conditions, the tilting pads themselves can become unstable in the form of shoe (pad) flutter, as mentioned previously.

All rotating machines vibrate when operating, but the failure of the bearings is mainly caused by their inability to resist cyclic stresses. The level of vibration a unit can tolerate is shown in the severity charts in [Figure 13-16](#). These charts are modified by many users to reflect the critical machines in which they would like to maintain much lower levels. Care must always be exercised when using these charts, since different machines have different size housings and rotors. Thus, the transmissibility of the signal will vary.

Thrust Bearings

The most important function of a thrust bearing is to resist the unbalanced force in a machine’s working fluid and to maintain the rotor in its position (within prescribed limits). A complete analysis of the thrust load must be conducted. As mentioned

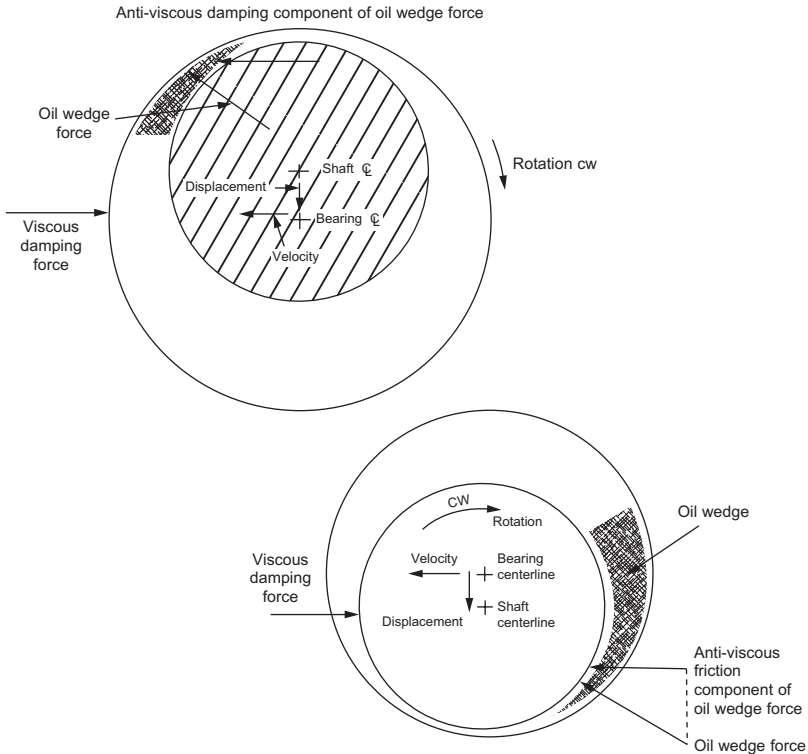


Figure 13-15 Oil whirl.

earlier, compressors with back-to-back rotors reduce this load greatly on thrust bearings. Figure 13-17 shows a number of thrust-bearing types. Plain, grooved thrust washers are rarely used with any continuous load, and their use tends to be confined to cases where the thrust load is of very short duration or possibly occurs at a standstill or low speed only. Occasionally, this type of bearing is used for light loads (less than 50 lb/in² [3.5 bar]), and in these circumstances the operation is probably hydrodynamic due to small distortions present in the nominally flat bearing surface.

When significant continuous loads have to be taken on a thrust washer, it is necessary to machine into the bearing surface a profile to generate a fluid film. This profile can be either a tapered wedge or occasionally a small step.

The tapered-land thrust bearing, when properly designed, can take and support a load equal to a tilting-pad thrust bearing. With perfect alignment, it can match the load of even a self-equalizing tilting-pad thrust bearing that pivots on the back of the pad along a radial line. For variable-speed operation, tilting-pad thrust bearings, as shown in Figure 13-18, are advantageous when compared to conventional taper-land bearings. The pads are free to pivot and form an angle for lubrication over a wide

speed range. The self-leveling feature equalizes individual pad loadings and reduces the sensitivity to shaft misalignments that may occur during service. The major drawback of this bearing type is that standard designs require more axial space than a non-equalizing thrust bearing.

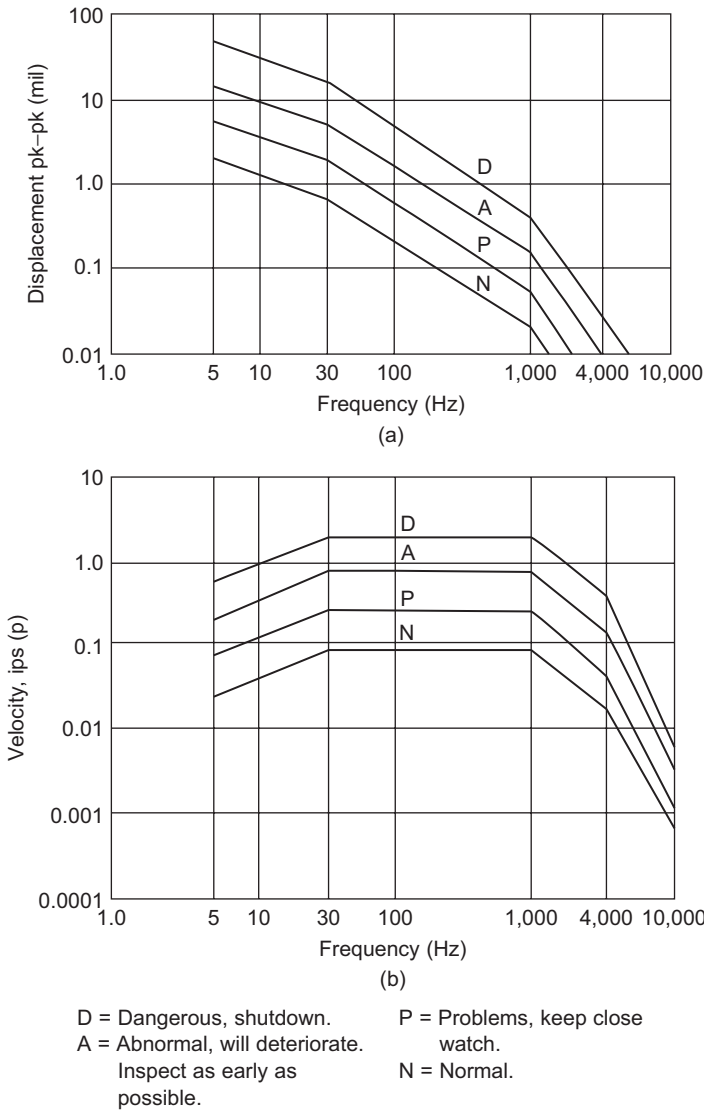


Figure 13-16 Severity charts: (a) displacement, (b) velocity. Severity chart: (c) acceleration.

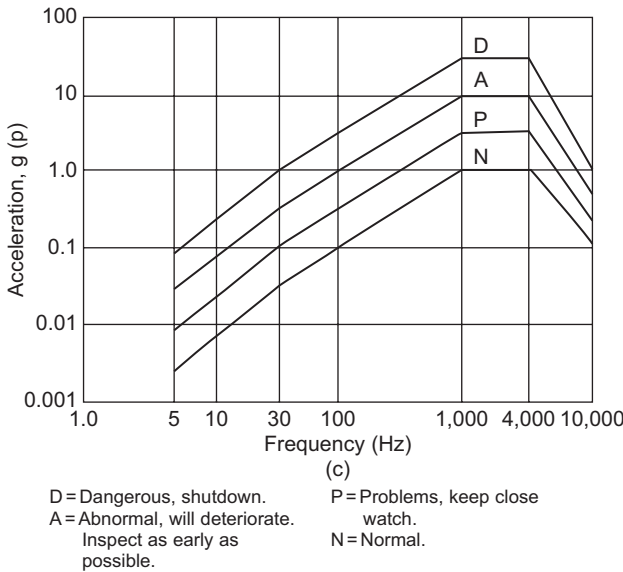


Figure 13-16 continued

Bearing type	Load capacity	Suitable direction of rotation	Tolerance of changing load/speed	Tolerance of misalignment	Space requirement
Plain washer	Poor		Good	Moderate	Compact
Taper land Bidirectional	Moderate		Poor	Poor	Compact
Unidirectional	Good		Poor	Poor	Compact
Tilting pad Bidirectional	Good		Good	Good	Greater
Unidirectional	Good		Good	Good	Greater

Figure 13-17 Comparison of thrust-bearing types.

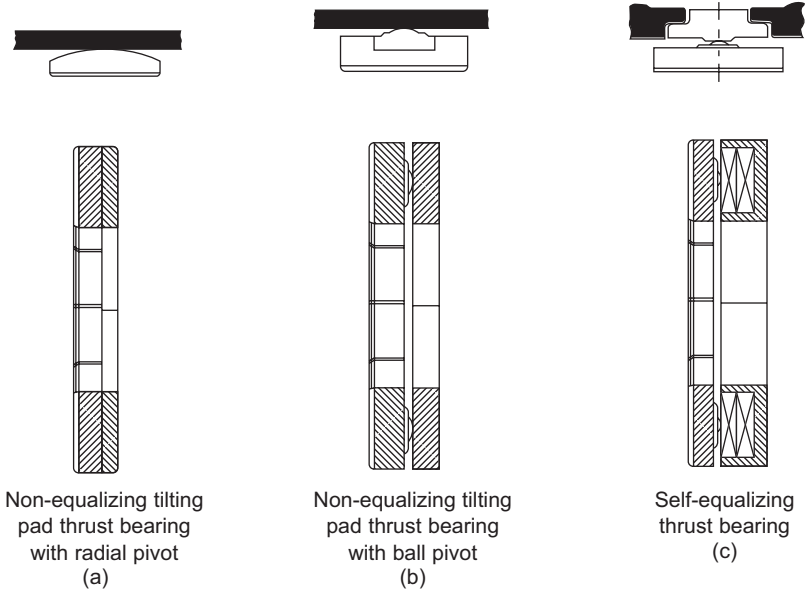


Figure 13-18 Various types of thrust-bearings.

Factors Affecting Thrust-Bearing Design

The principal function of a thrust bearing is to resist the thrust unbalance developed within the working elements of a turbomachine and to maintain the rotor position within tolerable limits.

After an accurate analysis has been made of the thrust load, the thrust bearing should be sized to support this load in the most efficient method possible. Many tests have proven that thrust bearings are limited in load capacity by the strength of the babbitt surface in the high load and temperature zone of the bearing. In normal steel-backed babbitted tilting-pad thrust bearings, this capacity is limited to between 250 and 500 psi (17 and 35 Bar) average pressure. It is the temperature accumulation at the surface and pad crowning that cause this limit.

The thrust-carrying capacity can be greatly improved by maintaining pad flatness and by removing heat from the loaded zone. By the use of high thermal conductivity backing materials with proper thickness and proper support, the maximum continuous thrust limit can be increased to 1,000 psi or more. This new limit can be used to increase either the factor of safety and improve the surge capacity of a given size bearing or reduce the thrust bearing size and consequently the losses generated for a given load.

Since the higher thermal conductivity material (copper or bronze) is a much better bearing material than the conventional steel backing, it is possible to reduce the babbitt thickness to .010–.030 of an inch (.254–.762 mm). Embedded thermocouples and RTDs will signal distress in the bearing if properly positioned. Temperature

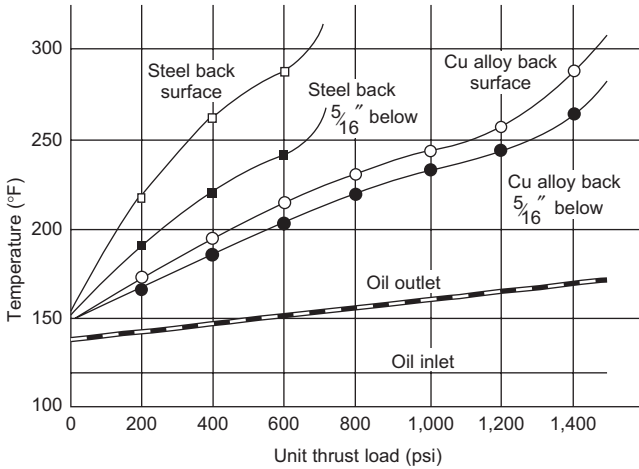


Figure 13-19 Thrust-bearing temperature characteristics.

monitoring systems have been found to be more accurate than axial position indicators, which tend to have linearity problems at high temperatures.

In a change from steel-backing to copper-backing a different set of temperature limiting criteria should be used. [Figure 13-19](#) shows a typical set of curves for the two backing materials. This chart also shows that drain oil temperature is a poor indicator of bearing operating conditions because there is very little change in drain oil temperature from low load to failure load.

Thrust-Bearing Power Loss

The power consumed by various thrust bearing types is an important consideration in any system. Power losses must be accurately predicted so that turbine efficiency can be computed and the oil supply system properly designed.

[Figure 13-20](#) shows the typical power consumption in thrust bearings as a function of unit speed. The total power loss is usually about 0.8–1.0% of the total rated power of the unit. New vectored lube bearings that are being tested show preliminary figures of reducing the power loss by as much as 30%.

Seals

Seals are very important and often critical components in turbomachinery, especially on high-pressure and high-speed equipment. This chapter covers the principal sealing systems used between the rotor and stator elements of turbomachinery. They fall into two main categories: (1) non-contacting seals, and (2) face seals.

Since these seals are an integral part of the rotor system, they affect the dynamic operating characteristics of the machine; for instance, both the stiffness and damping

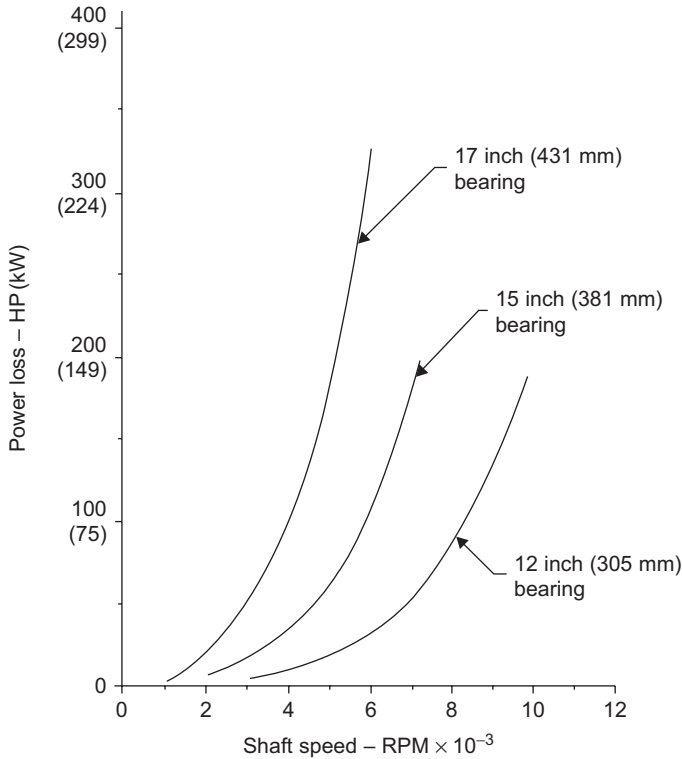


Figure 13-20 Total power loss in thrust bearings.

factors will be changed by seal geometry and pressures. Hence, these effects must be carefully evaluated and factored in during the design of the seal system.

Non-contacting Seals

These seals are used extensively in high-speed turbomachinery and have good mechanical reliability. They are not positive sealing. There are two types of non-contacting seals (or clearance seals): labyrinth seals and ring seals.

Labyrinth Seals

The labyrinth is one of the simplest of sealing devices. It consists of a series of circumferential strips of metal extending from the shaft or from the bore of the shaft housing to form a cascade of annular orifices. Labyrinth seal leakage is greater than that of clearance bushings, contact seals, or film-riding seals. Consequently, labyrinth seals are utilized when a small loss in efficiency can be tolerated. They are sometimes a valuable adjunct to the primary seal.

In large gas turbines labyrinth seals are used in static as well as dynamic applications. The essentially static function occurs where the casing parts must remain unjoined to allow for differences in thermal expansion. At this junction location, the labyrinth minimizes leakage. Dynamic labyrinth applications for both turbines and compressors are inter-stage seals, shroud seals, balance pistons, and end seals.

The major advantages of labyrinth seals are: their simplicity; reliability, tolerance to dirt, system adaptability, very low shaft power consumption, material selection flexibility, minimal effect on rotor dynamics, back diffusion reduction, integration of pressure, lack of pressure limitations, and tolerance to gross thermal variations. The major disadvantages are the high leakage, loss of machine efficiency, increased buffering costs, tolerance to ingestion of particulates with resulting damage to other critical items such as bearings, the possibility of the cavity clogging due to low gas velocities or back diffusion, and the inability to provide a simple seal system that meets OSHA or EPA standards. Because of some of the foregoing disadvantages, many machines are being converted to other types of seals.

Labyrinth seals are simple to manufacture and can be made from conventional materials. Early designs of labyrinth seals used knife-edge seals and relatively large chambers or pockets between the knives. These relatively long knives are easily subject to damage. The modern, more functional, and more reliable labyrinth seals consist of sturdy, closely spaced lands. Some labyrinth seals are shown in Figure 13-21. Figure 13-21 (a) is the simplest form of the seal. Figure 13-21 (b) shows a grooved seal is more difficult to manufacture but produces a tighter seal. Figures 13-21 (c) and 13-21 (d) are rotating labyrinth-type seals. Figure 13-21 (e) shows a simple labyrinth seal with a buffered gas for which pressure must be maintained above the process gas pressure and the outlet pressure (which can be greater than or less than the atmospheric pressure). The buffered gas produces a fluid barrier to the process gas. The eductor sucks gas from the vent near the atmospheric end. Figure 13-21 (f) shows a buffered, stepped labyrinth. The step labyrinth gives a tighter seal. The matching stationary seal is usually manufactured from soft materials such as babbitt or bronze, while the stationary or rotating labyrinth lands are made from steel. This composition enables the seal to be assembled with minimal clearance. The lands can therefore cut into the softer materials to provide the necessary running clearances for adjusting to the dynamic excursions of the rotor.

To maintain maximum sealing efficiency, it is essential that the labyrinth lands maintain sharp edges in the direction of the flow. This requirement is similar to that in orifice plates. A sharp edge provides for maximum vena contracta effect, and hence maximum restriction for the leakage flows (Figure 13-22).

High fluid velocities are generated at the throats of the constrictions, and the kinetic energy is then dissipated by turbulence in the chamber beyond each throat. Thus, the labyrinth is a device wherein there is a multiple loss of velocity head. With a straight labyrinth, there is some velocity carry-over that results in a loss of effectiveness, especially if the throats are closely spaced. To maximize the aerodynamic blockage effect of this carry-over, the diameters can be stepped or staggered to cause impingement of the expanding orifice jet on a solid, transverse surface. The leakage is approximately inversely proportional to the square root of the number of labyrinth lands. Thus, if

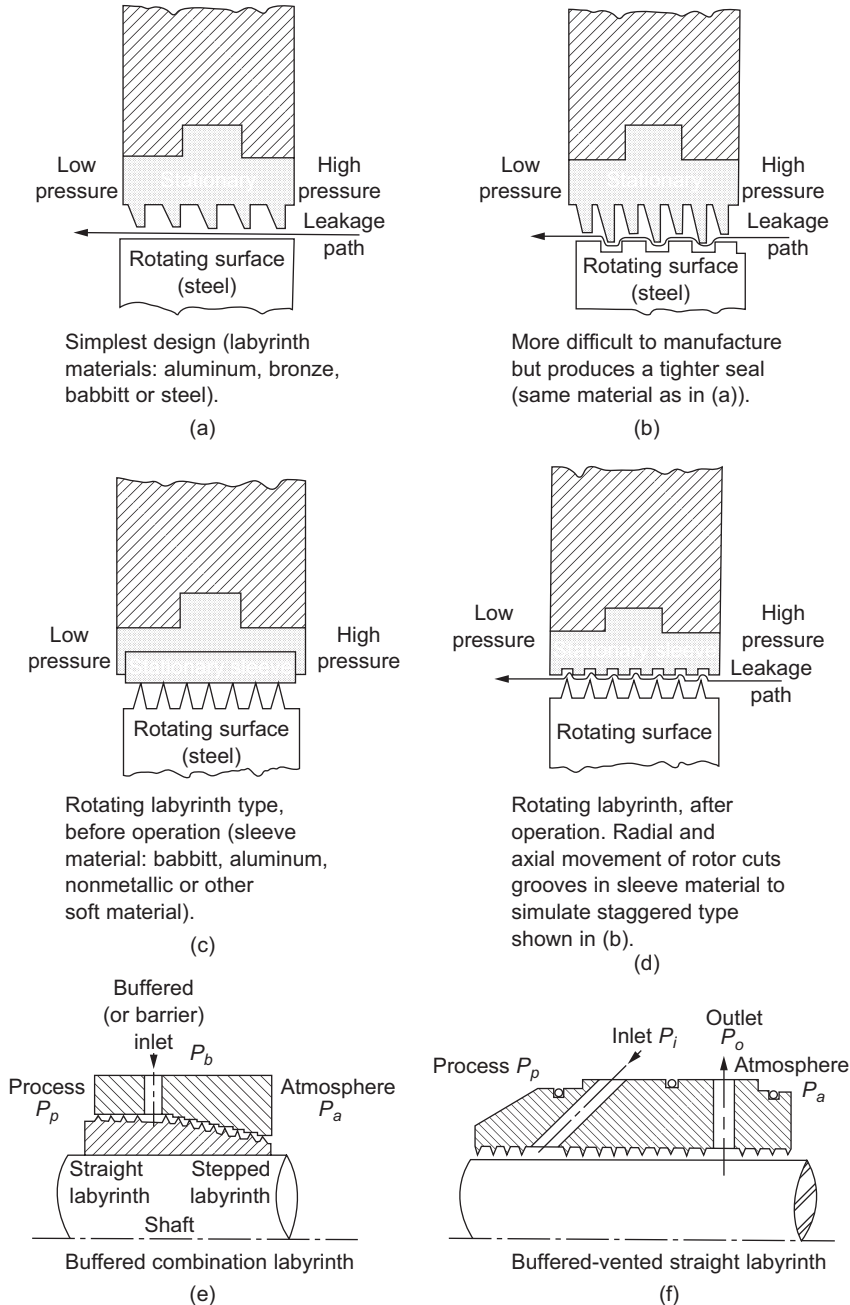


Figure 13-21 Various configurations of labyrinth seals.

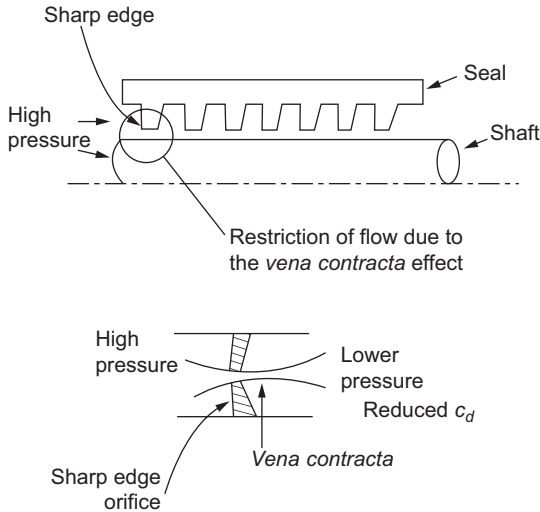


Figure 13-22 Theory behind the knife-edge arrangement.

leakage is to be cut in half at a four-point labyrinth, the number of lands would have to be increased to 16. The Elgi leakage formulae can be modified and written as:

$$\dot{m}_1 = 0.9A \left[\frac{\frac{g}{V_o} (P_o - P_n)}{n + \ln \frac{P_n}{P_o}} \right]^{1/2}$$

For staggered labyrinths, the equation can be written as:

$$\dot{m}_1 = 0.75A \left[\frac{\frac{g}{V_o} (P_o - P_n)}{n + \ln \frac{P_n}{P_o}} \right]^{1/2}$$

where

$\dot{m}_1$ = leakage, lb/sec

A = leakage area of single throttling, sq ft

P_o = absolute pressure before the labyrinth, lb/sq ft

V_o = specific volume before the labyrinth, cu ft/lbm

P_n = absolute pressure after the labyrinth, lb_f/sq ft

n = number of lands

The leakage of a labyrinth seal can be kept to a minimum by providing: (1) minimum clearance between the seal lands and the seal sleeve, (2) sharp edges on the lands to reduce the flow discharge coefficient, and (3) grooves or steps in the flow path for reducing dynamic head carry-over from stage to stage.

The labyrinth sleeve can be flexibly mounted to permit radial motion for self-aligning effects. In practice, a radial clearance of under 0.008 is difficult to achieve, except with very small high-precision machines. On larger turbines, clearances of 0.015–0.02 are generally used. During machine construction, it is important to measure and record these clearances because mechanical seizure or loss in aerodynamic efficiency can often be traced to incorrect labyrinth seal clearances.

The windback seal closely resembles the labyrinth but has an entirely different operational principle. A screw-thread device winds the oil, which is carried around the bore by the windage of the shaft, into an internal drain for return to the system as shown in Figure 13-23 (a).

Windback structures are extremely simple. Clearances about the shaft are ample, and the device has high reliability. When shaft speeds extend into the low regions where windage contributions are inadequate for effective operation, augmentation of windage can be achieved by special configurations of the shaft surface. Windbacks are also used as adjuncts to other types of seals, as shown in Figure 13-23 (b). With circumferential seals, windbacks can be used to keep oil splash from reaching the seal carbons when coking problems exist. In oil-buffered seals for compressors they are used to direct the small internal leakage into a pressurized drain to effect practically complete recovery of the leakage.

Ring (Bushing) Seals

The restrictive ring seal is essentially a series of sleeves in which the bores form a small clearance around the shaft. Thus, leakage is limited by the flow resistance in the restricted area and controlled by the laminar or turbulent friction. The API 617 codes characterize this type of seal. Most of the restrictive-type seals are of the floating type rather than the fixed type. The floating rings permit a much smaller leakage, and they can be of either the segmented type as shown in Figure 13-24 (a) or the rigid type as shown in Figure 13-24 (b).

Because of the minimal contact between the stationary ring and the rotor, these seals, when properly designed, are ideal for high-speed rotating machinery.

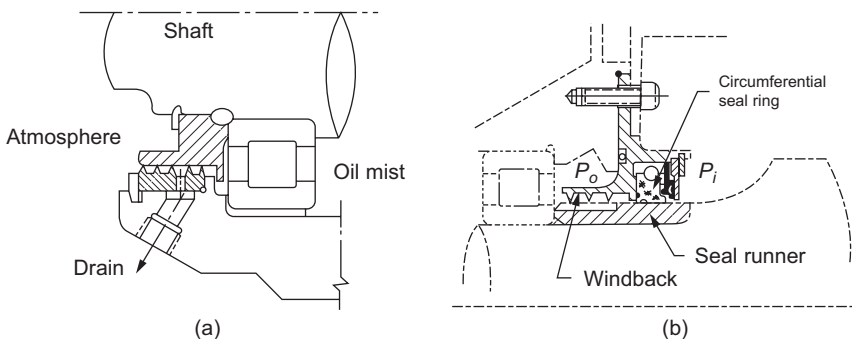


Figure 13-23 Windback seal.

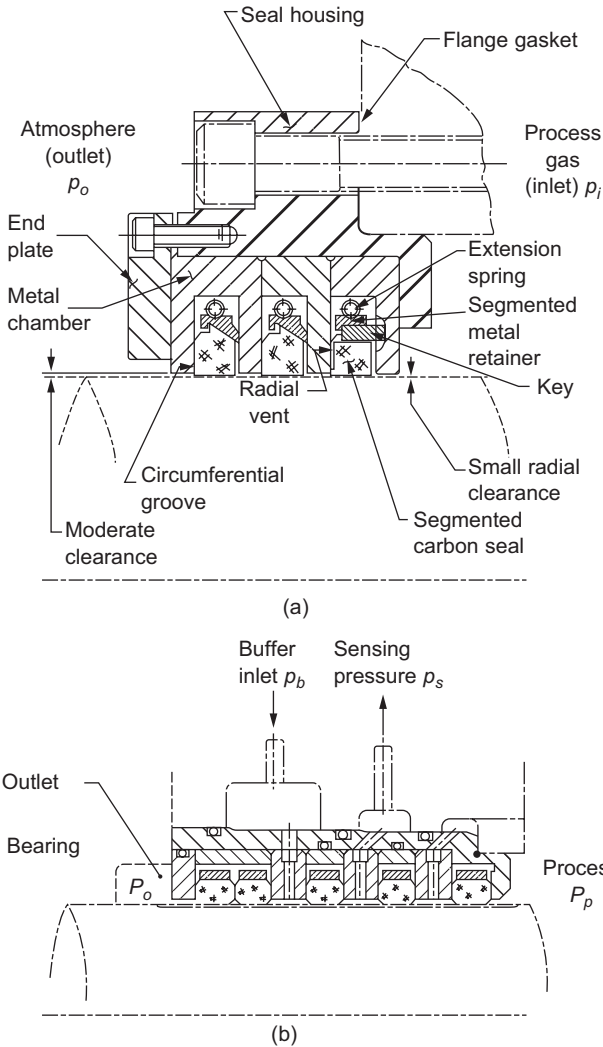


Figure 13-24 Floating-type restrictive ring seal.

When adequate lubrication and cooling fluid is available, the seal ring, manufactured from babbitt-lined steel, bronze, or carbon, will function satisfactorily. When the medium to be sealed is air or gas, carbon seal rings must be used. Carbon has self-lubricating properties. Cooling of the seal is provided by the leakage flow through the seal. Depending on the operating temperature and environment, aluminum alloys and silver are also used in the manufacture of the seal rings. Leakage limitation depends upon the type of flow and type of bushing. There are four types of flow: compressible and incompressible, each of which may be either laminar or turbulent. Ring seals are divided into two categories: fixed breakdown rings and floating breakdown rings, according to whether or not they are fixed with respect to the stationary housing.

Fixed seal rings

The fixed seal ring consists of a long sleeve affixed to a housing in which the shaft rotates with small clearance. It is an inexpensive assembly. However, since it is fixed, the seal behaves like a redundant bearing when rubbing occurs and, like the labyrinth, requires large clearances. Therefore, long assemblies must be used to keep leakage within reasonable limits. Since long seal assemblies aggravate alignment and rubbing problems, sturdier shafts are required to keep operating speeds in a subcritical region. The fixed-bushing seal almost always operates with appreciable eccentricity. This, plus the combination of a large clearance and a large eccentricity ratio, produces large leakages per unit length. Fixed-seal rings are therefore impractical where leakage is undesirable.

Floating seal rings

Clearance seals, which are free to move in a radial direction relative to the shaft and machine housing, are known as floating seals. These seals have advantages that very close annular clearance-type seals do not possess. The floating characteristic permits them to move freely with shaft motions and deflections, thereby avoiding the effects of severe rubbing.

Differential thermal expansion is a problem at high temperatures where the shaft and bushing are of dissimilar materials, or where there is any substantial temperature gradient between them. For example, the grades of carbon used commonly have a linear thermal expansion coefficient of one-third to one-fifth that of steel, necessitating the design of thermal expansion control into the carbon bushing. This is achieved by shrinking the carbon into a metallic retaining ring with a coefficient of expansion that equals or exceeds that of the shaft material.

It is good practice in critical applications to use bushings of a material with a slightly higher coefficient of thermal expansion than that of the shaft. Here, incipient seizure causes the bushings to grow away from the shaft. The large torque associated with high shearing intensity may necessitate locking the bushings against rotation if the unbalanced pressure forces seating them against the housing walls are insufficient to prevent rotation.

Build up of dirt or other foreign material between the seal ring and seat will result in damage to the journal and excessive seal spin on a floating seal ring unit. Soft materials, such as babbitt and silver, are notorious for trapping contaminants and causing shaft damage.

Mechanical (Face) Seals

This device forms a running seal between flat precision-finished surfaces. Its primary function is to prevent leakage. When used on rotating shafts, the sealing surfaces are in a plane perpendicular to the shaft, and the forces that hold the contact faces together will consequently be parallel to the shaft axis. For a seal to work properly, four sealing points must function as shown in [Figure 13-25](#). They are: (1) the stuffing-box face must be sealed, (2) leakage down the shaft must be sealed, (3) the mating ring in the

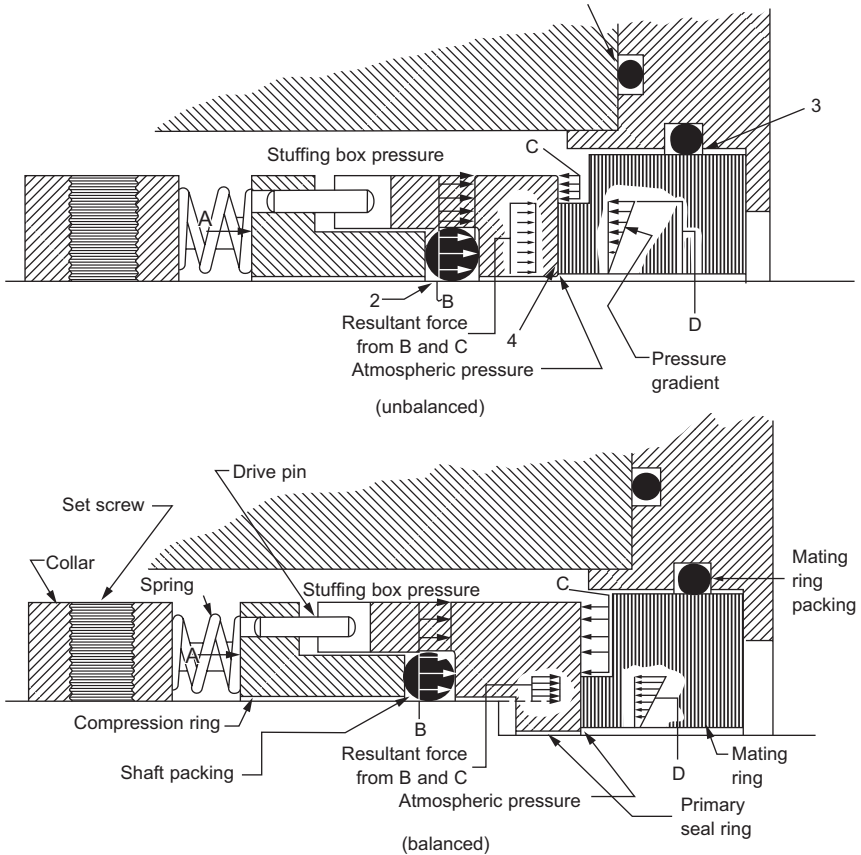


Figure 13-25 Unbalanced seal and balanced seal with step in shaft.

gland plate must be sealed in a floating design, and (4) the dynamic faces (rotary to stationary) must seal. Basically, most mechanical seals have the following components:

1. Rotating seal ring
2. Stationary seal ring
3. Spring devices to provide pressure
4. Static seals

A complete seal has two basic units: the seal head unit and the seal seat. The seal head unit consists of the housing, the end-face member, and the spring assembly. The seal seat is the mating member that completes the precision-lapped face combination that provides the seal.

The seal head may either rotate or remain stationary (attached to the body). Either one (head or seat) may rotate, while the other remains stationary. The movement of the sealing action depends on the direction of the pressure. This is illustrated in [Figure 13-26](#), which shows rotating and stationary heads.

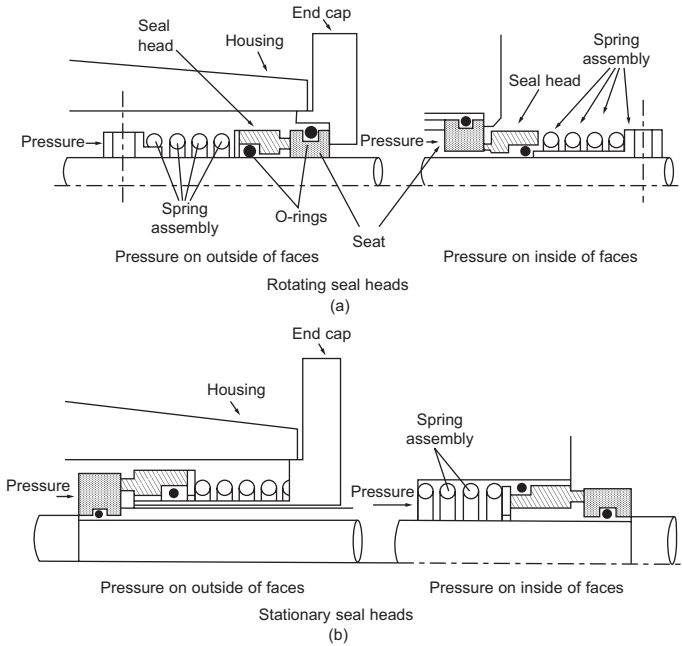


Figure 13-26 Rotating and stationary seal heads.

Some form of mechanical loading device, usually a spring, is needed to ensure that, in the event of a loss of hydraulic pressure, the sealing surfaces are kept closed. The amount of load on the sealing area is determined by the degree of “seal balance.” [Figure 13-27](#) shows what is meant by seal balance. A completely balanced combination occurs when the only force exerted on the sealing surfaces is the spring force, i.e., hydraulic pressure does not act on the sealing surfaces. The kind of spring that should be used depends upon a variety of factors: the space available, the loading characteristics required, the environment in which the seal is to operate, amongst others. Based on these considerations, either a single-spring or a multiple-spring design can be utilized. When a very small axial space is available, belleville springs, finger washers, or curved washers may be used.

A somewhat recent development is the use of magnetic force to obtain a face-loading action. Magnetic seals have provided reliable service under a variety of fluids and severe operating conditions. Some of the design advantages claimed are that magnetic seals are compact and lighter, provide an even distribution of sealing force, and are easy to assemble. [Figure 13-28](#) shows a simple magnetic seal.

Shaft sealing elements can be split up into two groups. The first may be called pusher-type seals and includes the O-ring, V-ring, U-cup, and wedge configurations. The second group are Bellow-type seals, which differ from the pusher-type seals in that they form a static seal between themselves and the shaft. [Figure 13-29](#) shows some typical pusher-type seals.

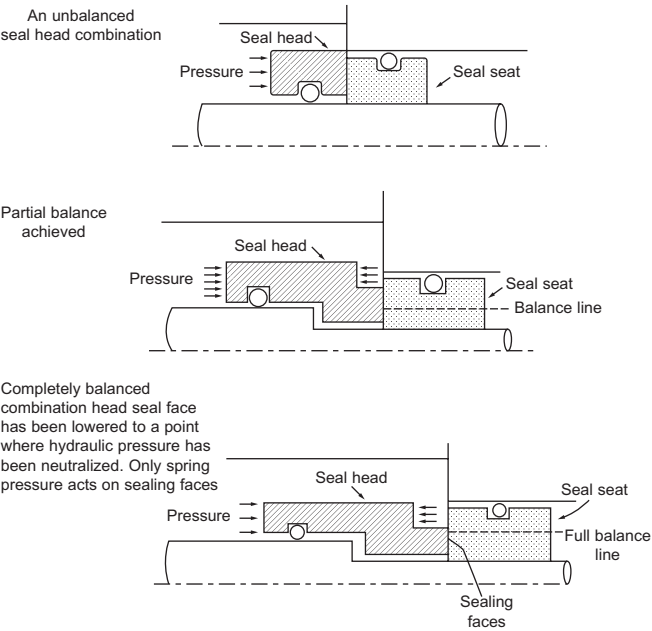


Figure 13-27 The seal balance concept.

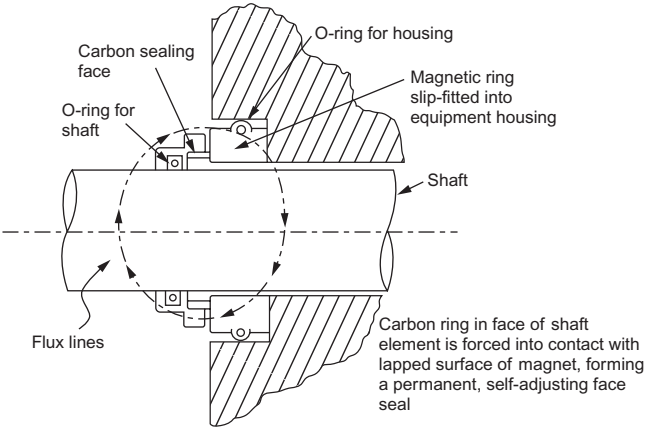


Figure 13-28 Simple magnetic-type seal.

A typical mechanical contact shaft seal has two major elements, as seen in [Figure 13-30](#). These are the oil-to-pressure-gas seal and the oil-to-uncontaminated-seal-oil-drain seal or breakdown bushing. This type of seal will normally have buffering via a single ported labyrinth located inboard of the seal and a positive shutdown device, which will attempt to maintain gas pressure in the casing when the compressor

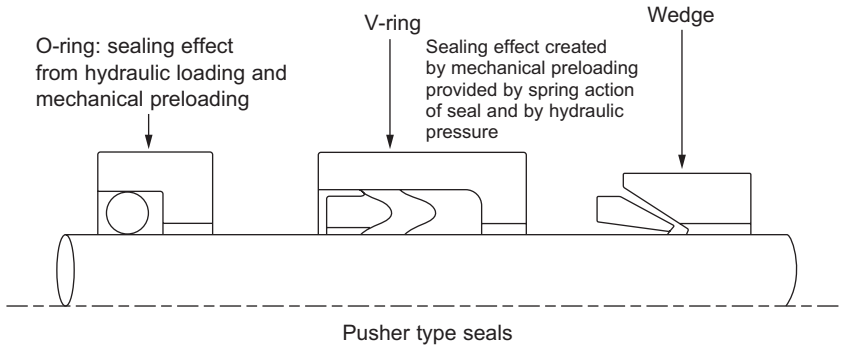


Figure 13-29 Various types of shaft sealing elements.

is at rest and seal oil is not being applied. For shutdown, the carbon ring is kept tightly sandwiched between the rotating seal ring and stationary sleeve with gas pressure to prevent gas from leaking out when no oil pressure is available.

In operation, seal oil pressure is held at a differential of 35–50 psi (2.4–3.5 Bar) over the process gas pressure which the seal is sealing against. This high-pressure oil can be seen entering in the top in [Figure 13-30](#) and completely fills the seal cavity. Some of the oil (a relatively small percentage, ranging from two to eight gpd per seal depending on machine size) is forced across the carbon ring seal faces, which are sandwiched between the rotating seal ring (rotating at shaft velocity) and the stationary sleeve (non-rotating and forced against the carbon ring by a series of peripheral springs). Therefore, the actual rotative speed of the carbon ring can be anywhere between zero rpm and full rotational speed. Oil crossing these seal faces contacts the process gas and is, thus, “contaminated oil.”

The majority of the oil flows out of the uncontaminated seal oil drain after taking a pressure drop from design seal oil pressure to atmospheric pressure across the breakdown bushing. An orifice is placed in parallel with the breakdown bushing to meter the proper amount of oil flow for cooling. The contaminated oil leaves through the drain to a degasifier for purification. The bearing oil drain can be either combined with the uncontaminated seal oil drain or kept separate; however, a separate system will increase bearing span and lower critical speeds.

Mechanical Seal Selection and Application

The following is a list of factors that have proven to be helpful in seal system design and selection:

1. Product
2. Seal environment
3. Seal arrangement
4. Equipment

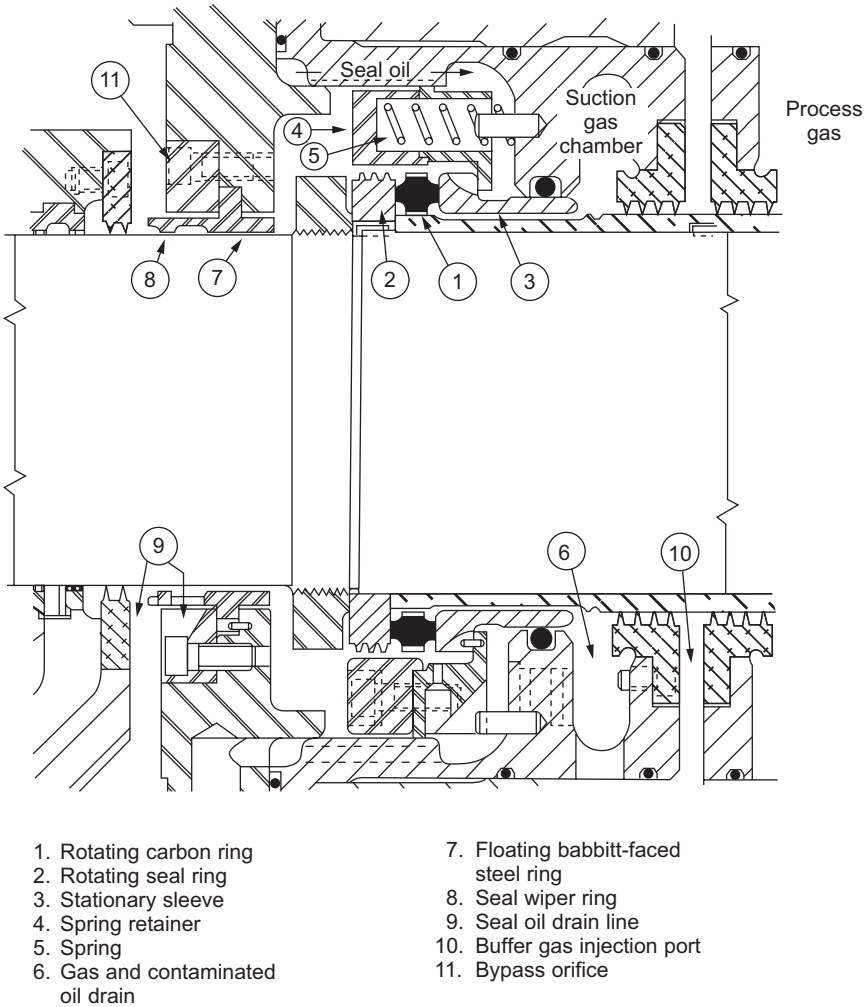


Figure 13-30 Mechanical contact shaft seal.

- 5. Secondary packing
- 6. Seal-face combinations
- 7. Seal gland plate
- 8. Main seal body

Product

The physical and chemical properties of the liquid being sealed will place constraints upon the type of seal arrangement, the materials of construction, and the seal design that can be used.

Pressure

The relative pressures of the material to be sealed affect the decision of whether to use a balanced or unbalanced seal design. Pressure also affects the choice of face material because of the seal-face loading.

If the service happens to be below atmospheric pressure, then special considerations are required to seal the material effectively. Most unbalanced seal designs are applicable up to 100 psig (7 Bar) stuffing-box pressure. At more than 100 psig (7 Bar), balanced seals should be used.

Seal manufacturers base their seal-face combination designs on *PV* ratings. These are the multiple of the face load (*P*) and the sliding velocity (*V*) of the faces. The maximum *PV* rating for an unbalanced seal is about 200,000 and about 2,250,000 for a balanced seal.

Temperature

The temperature of the liquid being pumped is important because it affects the seal-face material selection as well as face wear life. This is primarily a result of changes in lubricity of the fluid with changes in temperature.

Common seal designs may handle fluid temperatures in the 0°F to +200°F (−17°C to 93°C) range. When temperatures are above the +200°F (93°C) range, special metal bellows seals may be used up to the +650°F (343°C) range. Low temperature (−100°F to 0°F) (−73°C to −17°C) also requires special arrangements, since most hydrocarbons have little lubricity in this range.

The most important consideration concerning temperature is to avoid operating close to a temperature which will allow flashing of the liquid. Mechanical seals work well on many liquids; they work poorly on most gases.

Lubricity

In any mechanical seal design there is rubbing motion between the dynamic seal faces. This rubbing motion is most often lubricated by the fluid being pumped. Therefore, the lubricity of the pumped liquid at the given operating temperature must be considered to determine if the chosen seal design and face combination will perform satisfactorily.

Most seal manufacturers limit the speed of their seals to 90 fps (27.4 mps) with good lubrication of the faces. This is primarily due to the centrifugal forces acting on the seal which tend to restrict its axial flexibility.

Abrasion

When evaluating the possibility of installing a seal in a liquid that has entrained solids, several factors must be considered. Is the seal constructed in such a way that the dynamic motion will be restricted by fouling of the seal parts? The seal arrangement that is usually preferred when abrasives are present is a flushed single inside type with a face combination of very hard material. However, factors such as toxicity or corrosiveness of the material may dictate that other arrangements be used.

Corrosion

When considering the corrosiveness of the material being pumped, one must determine what metals will be acceptable for the seal body, what spring material may be used, what face material will be compatible with the liquid being pumped (that is, whether the binder or the carbon or tungsten carbide will be attacked, or whether the base metal of the plated seal face will be attacked), and what type of elastomer or gasket material can be used. The corrosion rate will affect the decision of whether to use a single- or multiple-spring design because the spring can tolerate a greater amount of corrosion without weakening it appreciably.

Toxicity

This factor is becoming an increasingly important consideration in the design of mechanical seals. Since the rubbing seal faces require liquid penetration to cool and lubricate them, it is reasonable to expect that there will be some vapor passing across the faces. This is in fact the case. A normal seal can be expected to “leak” from a few ppm to 10 cc/min. It is also generally accepted that the seal leakage rate will increase with speed.

Additional Product Considerations

1. Is the product thermosensitive? The heat generated by the seal faces may cause polymerization.
2. Is the product shear sensitive, i.e., will it harden due to turbulence?
3. If the product is highly flammable, be aware of possible ignition sources.
4. In-hazardous services plan for personnel protection in the event of seal leakage.
5. Products with dissolved gas must be properly vented. In most cases, vent the stuffing box back to the pump action.
6. Seals in cold services are extremely sensitive to moisture. There must be a way to “dry out the system” after repair.
7. Consideration must be given to the pressure and temperature that the seal will see during normal operation, start-up, shutdown, and upset conditions.
8. Vapor pressure of the product must be known to prevent vaporization in the stuffing box.

Seal Environment

Once an adequate definition of the product is made, the design of the seal environment can be selected. There are four general parameters that an environmental system may regulate or change:

1. Pressure control
2. Temperature control
3. Fluid replacement
4. Atmospheric air elimination

The most common environmental control systems include flushing, barrier fluids, quenching, and heating/cooling systems. Each has its use in regulating the parameters mentioned previously.

Seal Arrangement Considerations

There are four considerations:

1. Double seals have been the standard with toxic and lethal products, but maintenance problems and the seal design contribute to poor reliability. The double face-to-face seal should be looked at more closely.
2. Do not use a double seal in dirty service – the inside seal will hang up.
3. The API standard is a good guide to the use of balanced and unbalanced seals. Application of a balanced seal at too low a pressure may encourage face lift-off.
4. The number of arrangements and auxiliary features are more than 100. Regardless of the seal vendor, the arrangement will generally determine success.

Equipment

Too few people consider the equipment with the seal selection. In most cases, poor equipment will give poor seal performance, regardless of the seal or arrangement chosen. Also, beware that different pumps with the same shaft diameter and TDH may present different sealing problems (note: these same considerations may be used for troubleshooting).

Secondary Packing

More emphasis should be placed on secondary packing than it receives, especially if these members involve Teflon. Most seal designs using an O-ring for shaft packing give similar performance. A wide variation in performance is seen between various seal vendor designs when Teflon shaft packing is used. Depending on the seal arrangement, there can be a difference in mating-ring (stationary) packing performance when Teflon is used.

Seal-Face Combinations

Choices of seal-face combinations have come a long way in the last eight to 10 years. Stellite is being phased out in petroleum and petrochemical seal applications. Better grades of ceramic are being offered as the standard material. The cost of tungsten carbide has decreased considerably. Relapping services for tungsten are available near most industrial areas. Silicon carbide is gaining a hold on the market, especially in abrasive services. The technology of manufacturing tungsten carbide in a composite or overlay arrangement is offered by all of the major seal manufacturers. The dynamics of seal faces are better understood today.

Seal Gland Plate

The seal gland plate is an item that is caught in between the pump vendor and the seal vendor. The pump vendors can furnish good, reasonably priced alloy glands, but they are also limited because the gland is cast and must fit several seal designs. There are also some glands furnished by pump vendors that can be easily distorted by bolting.

Special glands requiring heating, quench, and drain with a floating-throat bushing on ANSI pumps should be furnished by the seal vendor. Gland designs on several ANSI pumps are not that impressive.

Main Seal Body

Designs differ considerably from one manufacturer to another. The term “seal body” makes reference to all rotating parts on a pusher seal, excluding shaft packing and the seal ring. The configuration or options offered on the seal body may be the chief reason to avoid the design for that particular service.

Seal Systems

In recent years, these systems have become more sophisticated to meet modern chemical process requirements and government restrictions. A simple seal system is the buffered and educted restrictive-ring seal system. This type of system, as shown in Figure 13-31, must operate with buffering pressure greater than the process and eductor pressure. The eductor pressure must in turn be below the atmospheric pressure. Problems with these systems are common because the eductor system does not have a large enough capacity, the buffered gas pressure is not higher than the process pressure, and in many cases the rings are installed backward.

The complex seal systems incorporate many different types of components to provide the most efficient sealing. Figure 13-32 shows a system that includes three different types of seals. The labyrinth seal initially provides the restriction that prevents the polymers contained in the process gas from clogging the seal rings. The labyrinth seal is followed by the two segmented circumferential contact seals and the four segmented restrictive-ring seals, which are primary seals in this combination. The primary restrictive-ring seals are followed by four circumferential segmented seal

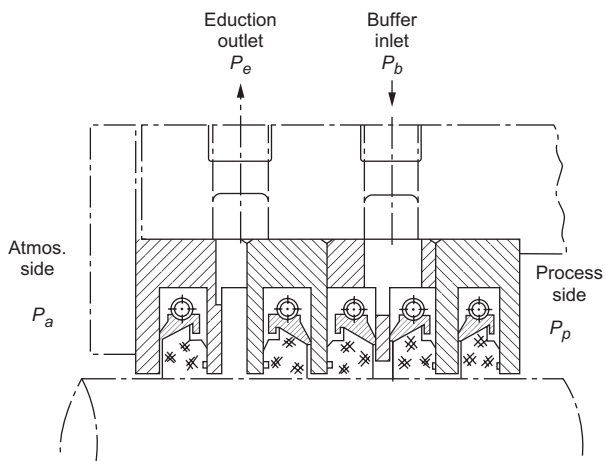


Figure 13-31 Restrictive ring seal system with both buffer and eduction cavities.

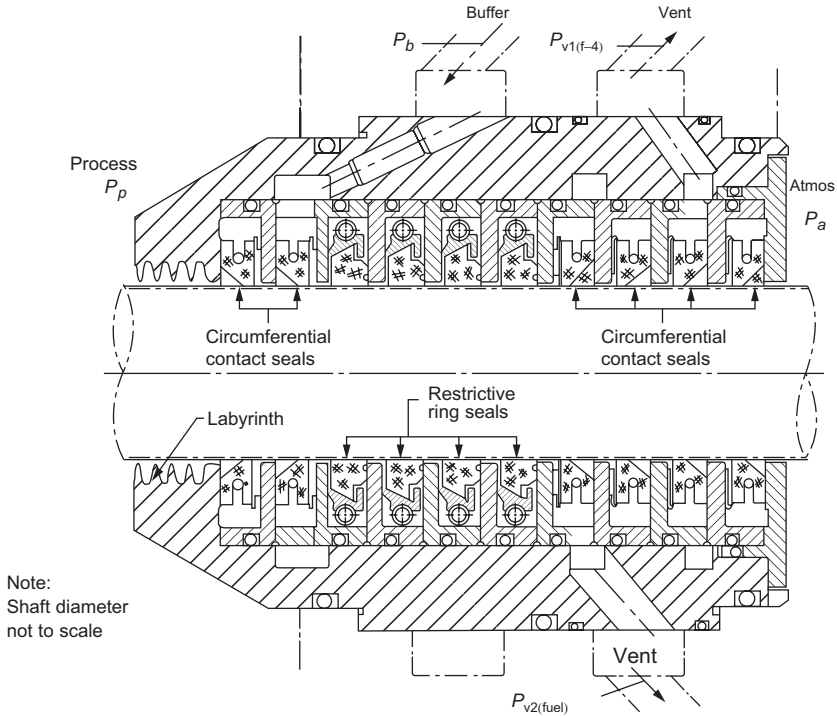


Figure 13-32 Multiple combination segmented gas seal system.

rings. A buffer gas is also introduced at the first set of circumferential contact seals, and an eductor is situated in the middle of the rear circumferential seals. Thus, this sealing system is very efficient in preventing any leakage and also for utilizing educted gas in the process.

Gas compressors operating on highly toxic or flammable gases may require redundant systems to assure no leakages. In many applications, such as refrigeration gas, buffer seals are required with the liquid-buffered face seal. A popular technique is to use a buffered labyrinth seal with a liquid seal.

Associated Oil System

One of the advantages of mechanical contact seals is that the associated seal oil supply system may be relatively simple compared to the system required with other types of seals, as seen in Figure 13-33. The relatively high oil-to-gas differential and wide allowable range allow simple differential regulators to be used to control the oil supply system rather than a complex overhead tank arrangement. The dark lines in Figure 13-34 represent the seal oil system used for this type of seal. Seal oil is taken from a controlled header “A” and dropped to the required ΔP via a relatively inexpensive regulator control. The sensing point for this ΔP control is off the contaminated drain cavity on the high-pressure end of the compressor. By sensing off the

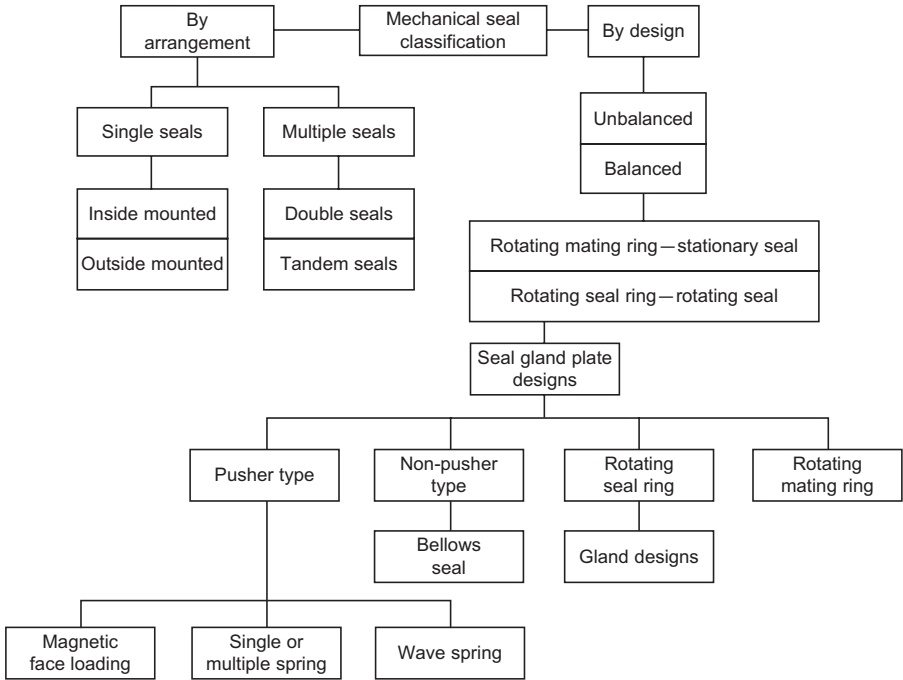


Figure 13-33 Mechanical seal classifications.

high-pressure end, a minimum of ΔP of oil to gas is always held on both ends of the compressor. Any pressurizing of the contaminated drain cavity due to buffer gas being used is also automatically followed by using a sensing point located in the contaminated drain oil cavity. In the system shown, the “uncontaminated oil” combines immediately with lube oil and returns to the reservoir where the “contaminated oil” can be trapped by a drainer and automatically drained to be optionally discarded or returned to the reservoir via a degassing tank.

Dry Gas Seals

The use of dry gas seals in process gas centrifugal compressors has increased over the last 30 years, replacing traditional oil film seals in most applications. Over 85% of centrifugal gas compressors manufactured today are equipped with dry gas seals.

Dry gas seals are basically mechanical face seals, consisting of a mating ring, which rotates and a primary ring, which is stationary. A cross-sectional view of a dry gas seal is shown in [Figure 13-35](#). The rotating assembly consists of the mating ring (with spiral grooves) mounted on a shaft sleeve held in place axially with a clamp sleeve and a locknut. It is typically pin driven. The mating ring with spiral grooves and the primary ring are held within the retainer assembly. The stationary assembly consists of the primary ring mounted in a retainer assembly held stationary within the compressor

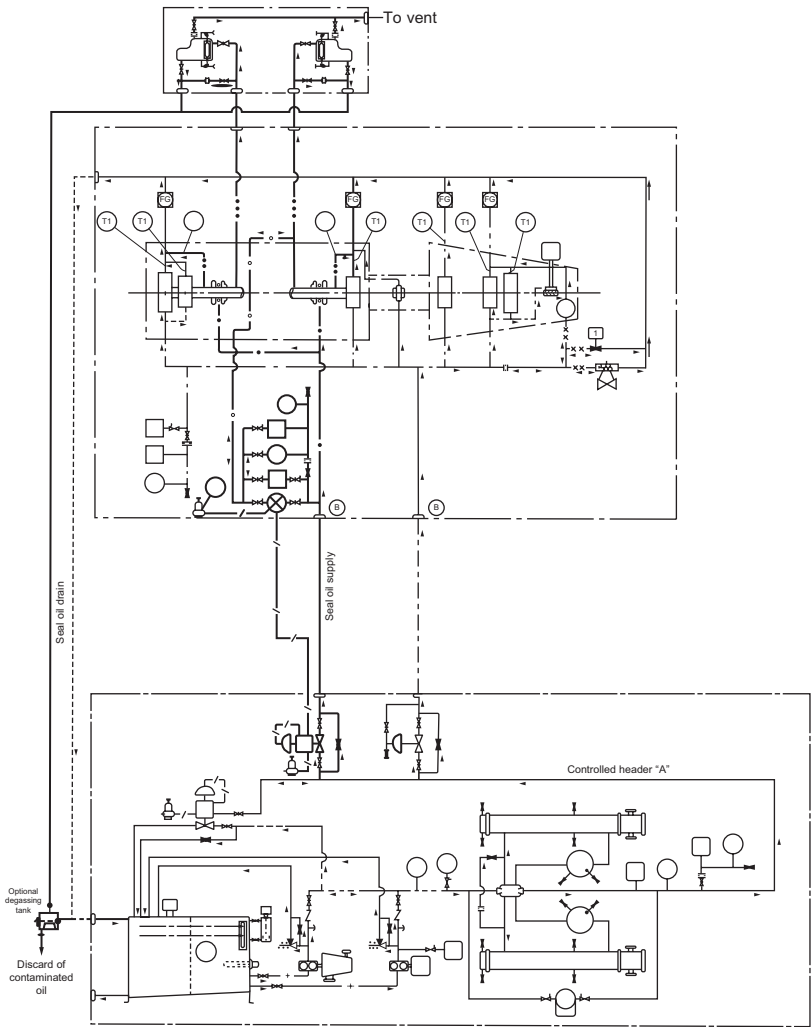


Figure 13-34 Mechanical contact seal and lube oil system.

housing. Under static conditions, the primary and mating rings are held in contact due to the spring load on the primary ring.

The spiral groove pattern, for a clockwise rotation, on the mating ring is shown in [Figure 13-36](#). The operating principle of the spiral grooved gas seal is that of a hydrostatic and hydrodynamic force balance. As gas enters the grooves, it is sheared towards the center. The sealing dam acts as a restriction to the gas outflow, thereby raising the pressure upstream of the dam. This increased pressure causes the flexibly mounted, primary ring to separate from the mating ring. During normal operation, the running gap is approximately three microns. Under pressurization, the forces exerted on the

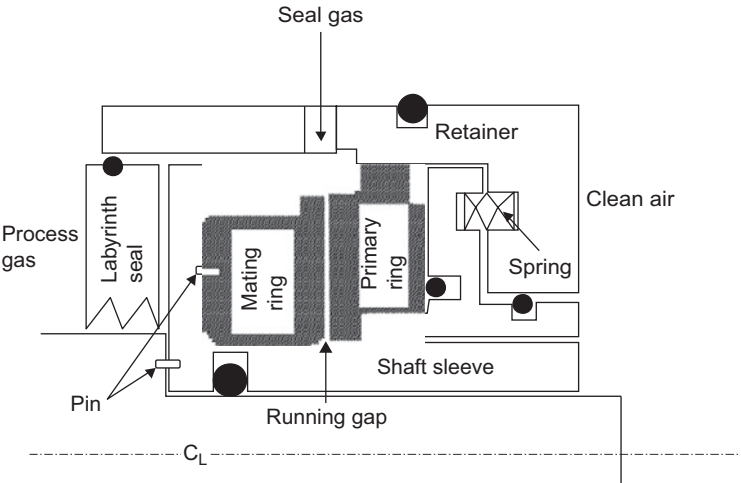


Figure 13-35 Single dry gas seal.

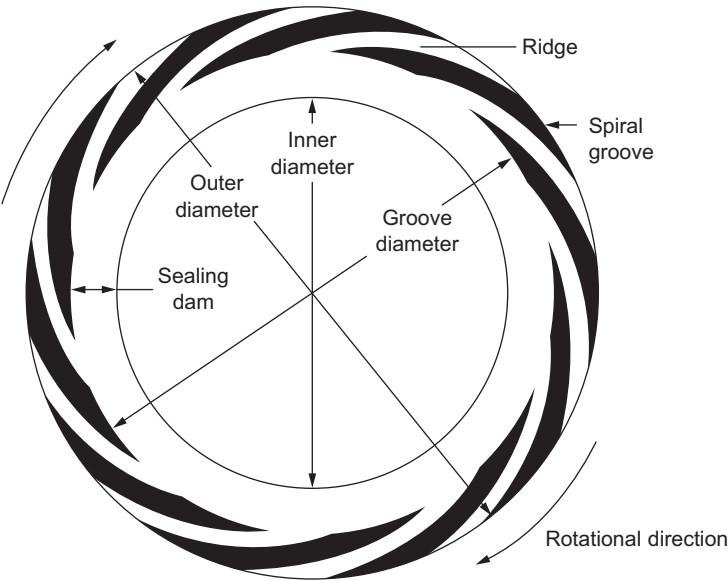


Figure 13-36 Spiral grooved mating ring (courtesy Proceedings Seventeenth Turbomachinery Symposium, Dry Gas Compressor Seals by Piyush Shah, John Crane Inc.).

seal are hydrostatic and are present whether the mating ring is stationary or rotating. Hydrodynamic forces are generated only upon rotation. The mating ring consisting of the logarithmic spiral grooves is the key to generating these hydrodynamic forces.

During operation, the grooves in the mating ring generate a hydrodynamic force that causes the primary ring to separate from the mating ring creating a “running gap” between the two rings, which effectively seals against the process gas. During normal operation, the running gap is approximately three microns. A sealing gas is injected into the seal, providing the working fluid, which establishes the running gap.

Tandem Dry Gas Seals

The tandem dry gas seal as seen in [Figure 13-37](#) relates to applications where small product leakages of process gas are admissible. The seal on the atmosphere side acts as a safety seal. The tandem arrangement provides a particularly high degree of operational safety. The seal on the product side and the seal on the atmosphere side are able to absorb the complete pressure differential. Under normal operating conditions the full pressure is reduced only by the seal on the product side. The space between the seal on the product side and the seal on the atmosphere side is cleared by a connection to the flare. The pressure differential to be sealed by the seal on the atmosphere side equals the flare pressure, so leakage to the atmosphere side or to the vent is very low. If the main seal fails, the second seal acts as a safety seal.

Tandem Dry Gas Seal with Labyrinth

A tandem seal with intermediate labyrinth is shown in [Figure 13-38](#). This seal is used in applications where the product leakages to the atmosphere as well as buffer gas leakages to the product are inadmissible, for example, on H₂, ethylene or propylene compressors. In this type of seal the product pressure to be sealed is reduced via the seal on the product side. The entire process gas leakage is discharged via connection

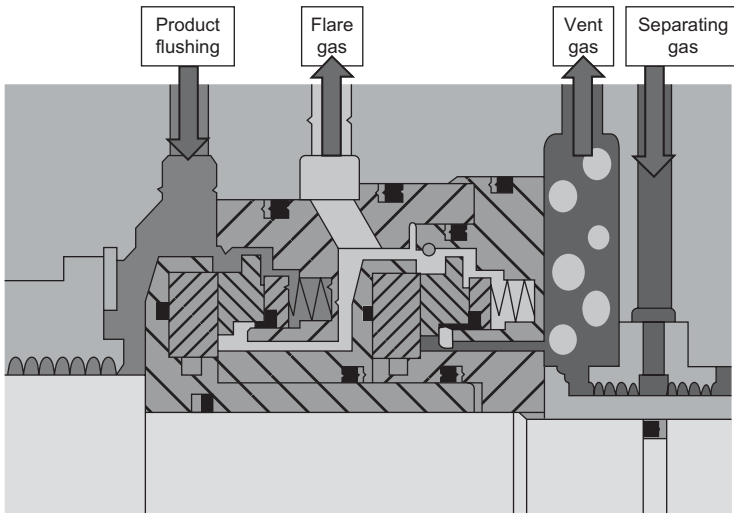


Figure 13-37 Tandem seal (courtesy Eagle Burgmann Germany GmbH & Co. KG).

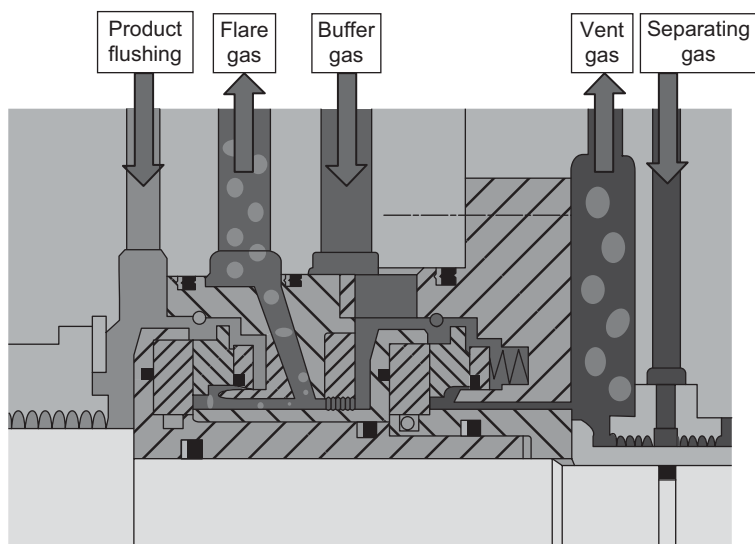


Figure 13-38 Tandem seal with intermediate labyrinth (courtesy Eagle Burgmann Germany GmbH & Co. KG).

to the flare. The seal on the atmosphere side is pressurized with a buffer gas such as nitrogen via connection to the buffer gas. The pressure of the buffer gas ensures that a current flows via the labyrinth to the primary vent outlet.

Double Gas Seals

A double gas seal, as shown in [Figure 13-39](#), is used where product leakages to the atmosphere are inadmissible and tandem arrangements are not suitable because of too small product gas pressures. Buffer gas leakages into the product must be admissible and the buffer gas pressure must be higher than the product pressure. This seal is used when a neutral buffer gas of suitable pressure is available. Typical applications are to be found mainly in the chemical industry, for example, on HC gas compressors. A buffer gas, for example, nitrogen, is fed between the seals at a higher pressure than the product pressure. One part of the buffer gas leakage escapes to the atmosphere side and the other part to the product side.

Operating Range of Dry Gas Seals

Gases ranging from inert gases such as nitrogen to highly toxic gaseous mixtures of natural gas and hydrogen sulfide can be sealed utilizing the optimum seal arrangements. The operating range of the spiral grooved dry gas seals is as follows:

Sealed Pressure: 2400 psi (165 Bar)
 Temperature: 500 °F (260 °C)
 Surface Speed: 500 ft/sec (152 m/sec)
 M.W.: 2–60

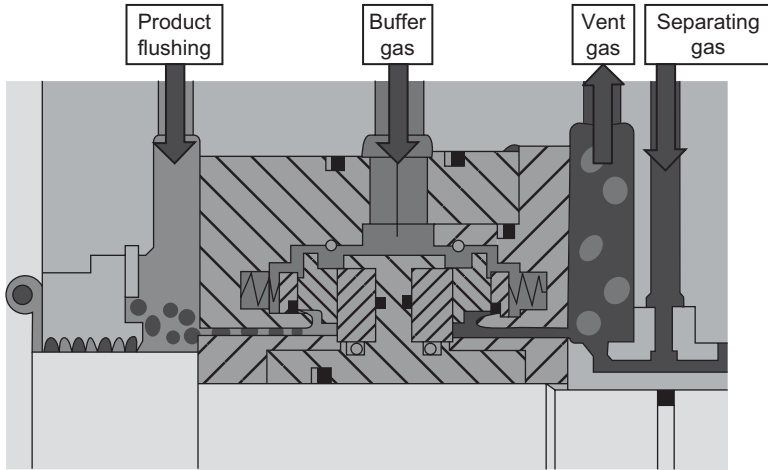


Figure 13-39 Double seal (courtesy Eagle Burgmann Germany GmbH & Co. KG).

Dry Gas Seal Materials

The gas composition, contaminants in the gas stream, operating temperatures, and process conditions dictate the choice of materials. The most common materials of construction are as follows:

- Mating Ring: Tungsten Carbide, Silicon Carbide
- Primary Ring: Carbon, Silicon Carbide
- O-Rings: Elastomers (Viton, “Kalrez”)
- Hardware: 300 or 400 series ss (Sleeves, discs, retainer rings)
- Coil Springs: 316 ss, Hastelloy

Dry Gas Seal Systems

The use of dry gas seals requires a system designed to supply sealing gas to the seal as a working fluid for the running gap. These gas seal systems are normally supplied by the compressor OEM mounted on the compressor base plate. There are two basic types of gas seal systems, differential pressure (ΔP) control and flow control. Differential control systems control the supply of seal gas to the seal by regulating the seal gas pressure to a pre-determined value typically 15 psi (1 Bar) above the sealing pressure. This is accomplished through the use of a differential pressure control valve. Flow control systems control the supply of seal gas to the seal by regulating the seal gas flow through an orifice upstream of the seal. This is accomplished through the use of a differential pressure control valve monitoring pressures on either side of the orifice.

Dry Gas Seal Degradation

Contamination of the seal by foreign objects leads to seal failures. The running gap between the primary and mating gas seal rings is typically around three microns.

Injection of any type of solids or liquids into this very narrow seal running gap can cause degradation of seal performance. This would create excessive gas leakage to the vent and eventual failure of the seal.

Since the typical operating gaps between the two sealing surfaces range from 0.0001 in to 0.0003 in, the resultant leakage is very small in magnitude. Under conditions of static pressurization beyond 50–75 psi (3.4–5.17 Bar), the seal leaks a very small amount. This leakage increases with increasing pressure and reduces with increasing temperature. Increased viscosity of gases at higher temperatures reduces the amount of seal leakage. For example, a 4-in (101.6 mm) shaft seal on a natural gas compressor statically pressurized to 1000 psi (69 bar) will leak about 1 scfm (0.03 scmm). Under dynamic condition, due to the pumping effect of the spiral grooves, the leakage increases as well.

The power loss can also be increased along with seal contamination. The seal surfaces being non-contacting under dynamic conditions the power loss associated with dry gas seals is very small. The power loss for a 10-in. (254 mm) seal operating at 1,000 psi (69 Bar) and 10,000 rpm is about 12–14 kW. With damaged seal surfaces, these losses can be increased by 20–30%.

Foreign material within the seal results in increased shearing forces between the primary and mating rings, causing overheating of the seal components, leading to O-ring extrusion or some other mechanical form of seal failure. The major areas from which gas seals contamination occurs are:

- Process gas leakages from the inboard or high-pressure side of the seal.
- Bearing lubrication oil from the outboard or low-pressure side of the seal.
- The seal gas injected into the seal being contaminated upstream of the injection.

Contamination from Process Gas

Contamination from process gas can occur when there is insufficient supply of sealing gas into the seal, allowing process gas to come into direct contact with the seal ring faces. Contaminants existing within the process gas can then damage the seal.

Contamination from Bearing Lubrication Oil

A barrier seal is required on the outboard side of the dry gas seal, between the gas seal and the compressor bearing. The primary function of the barrier seal, typically buffered with air or nitrogen, is to prohibit the flow of bearing lubrication oil into the gas seal. Contamination of the dry gas seal from lube oil can occur when the barrier seal fails to function as intended.

Contamination from Seal Gas Supply

Contamination from the seal gas supply occurs when the sealing gas is not properly treated upstream of the dry gas seal. Gas seal manufacturers have stringent requirements for seal gas quality. Typically, the sealing gas must be dry and filtered of particles three microns and larger. Filters are normally provided in the gas seal system to meet this requirement.

Dry gas seals operate under extremely tight tolerances, which demand that special care be taken in the design of the gas seal environment, and in the operation of the compressor and gas seal system. While the threat of seal degradation and reduced seal life due to outside influences is real, the detrimental effects of these factors can be minimized.

The replacement of mechanical seals by dry gas seals must be closely examined. There have been many cases where the replacement has caused the compressor to operate in an unstable manner. This is due to the fact that removal of the mechanical seal causes a change in the damping of the rotor and can cause the rotor to operate closer to its critical speed.

Bibliography

- Abramovitz, S., "Fluid Film Bearings, Fundamentals and Design Criteria, and Pitfalls," Proceedings of the 6th Turbomachinery Symposium, December 1977, pp. 189–204.
- Childs, D.W., and Vance, J.M., "Annular Gas Seals and Rotordynamics of Compressors and Turbines," Proceedings of the 26th Turbomachinery Symposium, Texas A&M University, 1997, p. 201.
- "Effects of Compressible Annular Seals," Proceedings of the 24th Turbomachinery Symposium, Texas A&M University, 1995, p. 175.
- Egli, T., "The Leakage of Steam through Labyrinth Seals," Trans. ASME, 1935, pp. 115–122.
- Evenson, R.S., Mason, B., Frederick, D.V., St. Onge, and Alain, G., "Development and Field Application of a Single Rotor Design Dry Gas Seal," Proceedings of the 24th Turbomachinery Symposium, Texas A&M University, 1995, p. 107.
- FAG "Rolling Bearing Damage," Publication No. WL 82 102/2 Esi, 1995.
- FAG "Rolling Bearings, Fundamentals, Types, Design," Publication No. WL 43 1190 EA, 1996.
- Garner, D.R., and Leopard, A.J., "Temperature Measurements in Fluid Film Bearings," Proceedings of the 13th Turbomachinery Symposium, Texas A&M University, 1984, p. 133.
- Herbage, B., "High Efficiency Fluid Film Thrust Bearings for Turbomachinery," 6th Proceedings of the Turbomachinery Symposium, Texas A&M University, December 1977, pp. 33–38.
- Herbage, B.S., "High Speed Journal and Thrust Bearing Design," Proceedings of the 1st Turbomachinery Symposium, Texas A&M University, October 1972, pp. 56–61.
- Jackson, C., "Radial and Thrust Bearing Practices with Case Histories," Proceedings of the 14th Turbomachinery Symposium, Texas A&M University, 1985, p. 73.
- King, T.L., and Capitaio, J.W., "Impact on Recent Tilting Pad Thrust Bearing Tests on Steam Turbine Design and Performance," Proceedings of the 4th Turbomachinery Symposium, Texas A&M University, October 1975, pp. 1–8.
- Leopard, A.J., "Principles of Fluid Film Bearing Design and Application," Proceedings of the 6th Turbomachinery Symposium, Texas A&M University, December 1977, pp. 207–230.
- Mayeux, T., Paul, Feltman Jr., and Paul, L., "Design Improvements Enhance Dry Gas Seal's Ability to Handle Reverse Pressurization," Proceedings of the 25th Turbomachinery Symposium, Texas A&M University, 1996, p. 149.
- Richards, R.L., Vance, J.M., Paquette, D.J., and Zeidan, F.Y., "Using a Damper Seal to Eliminate Subsynchronous Vibrations in Three Back-to-Back Compressors," Proceedings of the 24th Turbomachinery Symposium, Texas A&M University, 1995, p. 59.

- Salamone, D.J., "Journal Bearing Design Types and Their Applications to Turbomachinery," Proceedings of the 13th Turbomachinery Symposium, Texas A&M University, 1984, p. 179.
- Scharrer, J.K., Pelletti, and Joseph, M., "Leakage and Rotordynamic Effects of Compressible Annular Seals," Proceedings of the 24th Turbomachinery Symposium, Texas A&M University, 1995, p. 175.
- Shah, P., "Dry Gas Compressor Seals," Proceedings of the 17th Turbomachinery Symposium, Texas A&M University, 1988, p. 133.
- Shapiro, W., and Colsher, R., "Dynamic Characteristics of Fluid Film Bearings," Proceedings of the 6th Turbomachinery Symposium, Texas A&M University, December 1977, pp. 39–53.
- Southcott, J.F., Sweeney, J.M., Feltnan Jr., and Paul, L., "Dry Gas Seal Retrofit," Proceedings of the 24th Turbomachinery Symposium, Texas A&M University, 1995, p. 221.
- Takeuchi, Takao, Kataoka, Tadashi, Nagasaka, Hiroshi, Kakutani, Momoko, Ito, Masanobu, Muraki, and Ryoji "Advanced Dry Gas Seal by the Dynamic Ion Beam Mixing Technique," Proceedings of the 27th Turbomachinery Symposium, Texas A&M University, 1998, p. 39.

14 Gears

Gearing is one of the most important components between prime movers and driven units. If gearing is not selected properly, it can cause many problems. Gearing transmits great power at high rotational speeds. Recent advances in turbomachinery technology, especially in turbines, compressors, couplings, and bearings, have required gearing to withstand high external forces. To design problem-free equipment, it is important to consider the effect of the external system on gearing. Thus, all the factors that influence design, application, and operation of gear drives should be considered in the design phase.

Since problems encountered with gears are complex, it is unfair to blame the gear manufacturers alone. The gear supplier is much less-informed about the package than any other group. Problems should be handled as a team effort between manufacturers and users. One factor causing problems is that the system is not timed in terms of spring constants and masses. The gear is usually the only item required to operate with metal parts in such close contact with other components. This setup can result in early failure. Gearing is also subjected to cyclic loading varying from 0 to 55,000 cycles/min.

With current materials and heat-treating techniques, the use of high-hardness gearing with tooth loads up to 5,000 psi (34,474 kPa) of face at pitch line velocities of 20,000–30,000 ft/min (6,096–9,144 m/min) is not at all uncommon. In turbine-driven test equipment, gear drives have been built with pitch line velocities as high as 55,000 ft/min (16,764 m/min) and rotational speeds approaching 100,000 rpm. The magnitude of internal forces and material stresses coupled with the high speeds has resulted in gear drives that are dynamically complicated and sensitive to influences from other components in the system.

The system characteristics of the entire train must be known, so that the selection of the gear will be proper. The major points affecting the system are as follows: (1) couplings, (2) external vibration, (3) operating conditions, (4) thrust loads, and (5) mounting type.

Couplings are a constant source of unbalance vibration, and critical speed changes can be attributed to spacer shift and wear. Coupling keys can also cause severe housing vibration, whereas shaft vibration can remain low. Therefore, it is important to monitor vibration with accelerometers in addition to proximity probes. Gear failure from high vibration is common when the gear and pinion teeth operate within a distance of few hundred microinches between each other. Accelerometers can also monitor gear

mesh frequencies and thus act as early warning devices. Operating conditions must be known in detail.

In many cases, the gear manufacturer is provided with only the design horsepower of the machine. Actual transmitted loads can be much higher due to the proximity of torsional or lateral critical speeds. Surge in centrifugal compressors can cause severe overload and result in failures.

External thrust loads are another major problem, and in many cases, they lead to double-helical gear selection as shown in [Figure 14-1](#). The gear housing and the mounting type of the gear train are very important considerations in the overall life of the unit, since improper mounting and expansion of the gear housing can lead to misalignment problems.

A substantial structure to support the gear drive weight, thrust, and torque reactions with minimum load deflections must be provided. At least two dowels for locating each gear housing are required, and it is necessary to minimize housing vibration from any source. Ideally, the structures should be reinforced concrete or steel filled with grout. The inclusion of oil reservoirs in the structure supporting major train components should be avoided, since unavoidable thermal changes will have adverse effects on the alignment. If a reinforced concrete or a filled structure cannot be provided,

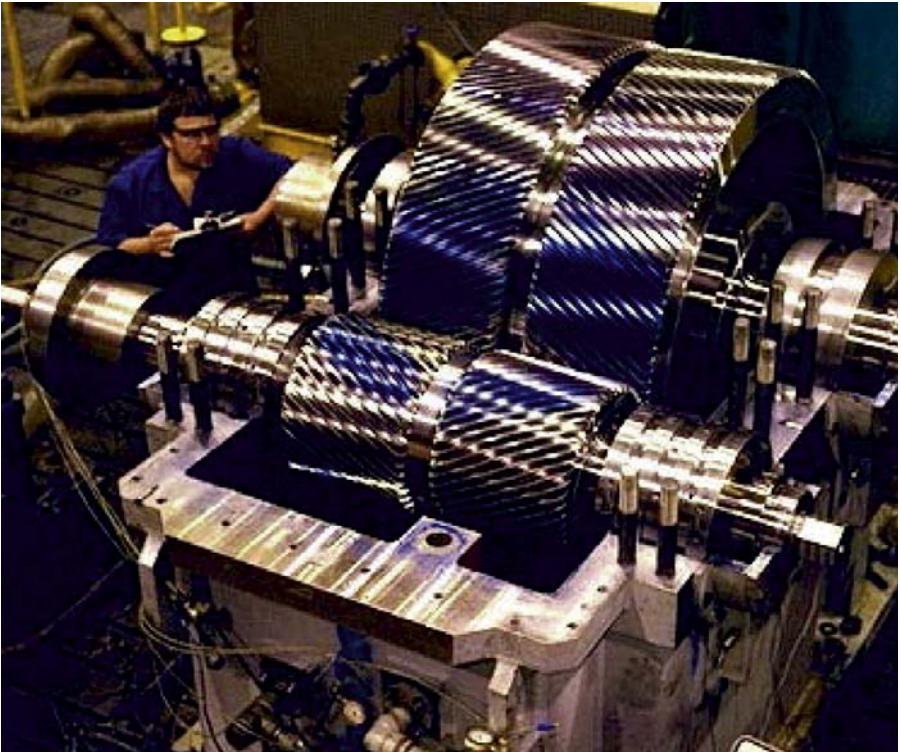


Figure 14-1 A double helical 54,000-HP (40 MW) gearbox (courtesy Lufkin Industries, Inc.).

resonance due to train component mass and structure stiffness at system rotational frequencies or harmonics should be avoided.

Gear Types

The choice between single- and double-helical gearing is sometimes difficult. Both gearing types can be made to equal limits of accuracy. Control of the gearing accuracy is only a function of the gear generating machines, which need maintenance, and operator skill for machining techniques. A hobbled gear is generated in a continuous process by a simple and easily maintained rack-form cutter, which will produce gearing of extremely high-profile accuracy with virtually immeasurable spacing errors and uniform lead. When both helices of a double-helical gear are cut at the same time, or sequentially without a change in setup, apex position error will be virtually unobservable in operation, and axial vibration excitation from the mesh will be negligible. The same basic equipment can be used for generating either single- or double-helical gears. Most gas turbine application gearing, whether single or double helical, is finished with a grinding process, to achieve the necessary tooth accuracy and surface finish.

Thrust loading is a significant problem in the design of any gear unit, and its effects differ based on the choice of single- or double-helical gearing. In either case, an accurate estimate of the thrust loading is required to make an intelligent compensation for it. With double-helical gearing, continuous axial loading can be accommodated by a slight increase in capacity to account for the helix load imbalance. American Petroleum Institute (API) 613 and 617 specifications require that single-helical gears must have a thrust bearing. It is also recommended (but not required) that double-helical gears must have thrust bearings.

The increase in cost and the reduction of efficiency thus caused is only a fraction of that incurred when a large diameter and high-velocity thrust bearing must be mounted on a single-helical pinion shaft.

Intermittent loading, such as that from gear couplings mounted on thermally expanding shafting, is a different problem. This problem is accommodated in a double-helical unit by judicious selection of helix angle and coupling size, so that axial coupling forces resulting from the transmission of torque are less than the thrust force produced by each helix of the gear. This design assures that the coupling will slip to relieve any axial loading and that the power balance on the two helices will be maintained.

High-speed gear couplings are selected in which the pitch diameter is substantially smaller than that of the pinion. Axial forces produced will be high, and the combination should be examined very closely as a potential trouble source. Diaphragm and disc couplings may help avoid a gear coupling-induced axial force.

Double-helical gearing is usually the first choice in providing the accuracy of loading and smoothness of operation. Predictable performance renders unnecessary complex deviations from easily defined and measured geometry. These gear sets will be more efficient and have unmatched reliability if properly applied. Reasonable

intelligence must be exercised in coupling selection. This discretion will assure vibration and noise levels that are often indistinguishable from that of the connected machinery.

In single-helical gearing, all forces externally generated must be added to the thrust produced by the gear itself, and the total force is used on each shaft to select the high-speed shaft thrust bearing. An error in thrust or bearing capacity estimation will result in frequent failures of the thrust bearing or the associated shafting. Single-helical gearing, due to asymmetrical loading from the helix, has two sources of design difficulty that do not exist in double-helical gear sets. The effective center of tooth pressure oscillates back and forth across the face, putting substantial alternating loads on the shaft bearings. This oscillation results in peak-bearing loads substantially greater than those calculated, which can lead to early bearing failure. In addition, the helix-induced thrust force causes the gearing to try to skew in the housing, unbalancing the bearing loading and forcing the gearing to run out of parallel. Crowning of single-helical gearing is used to counter the effects of shaft misalignment.

Factors Affecting Gear Design

A transverse section through a gear and pinion mesh is shown in Figure 14-2 with some of the major points in gear and pinion interaction. Figure 14-3 shows the terminology used to describe helical gears. The major factors affecting gear performance are as follows: (1) pressure angle, (2) helix angle, (3) tooth hardness, (4) scuffing

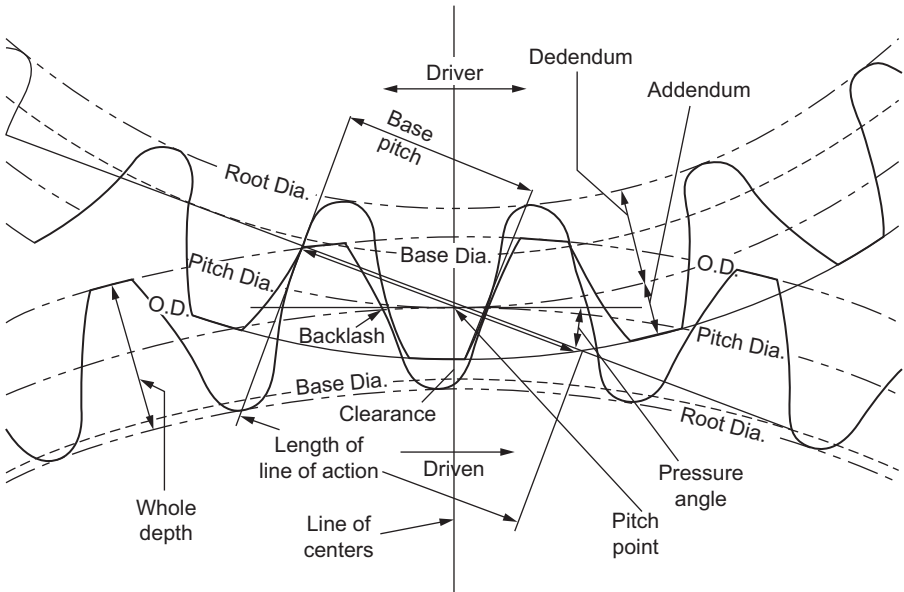


Figure 14-2 Transverse section through gear and pinion mesh.

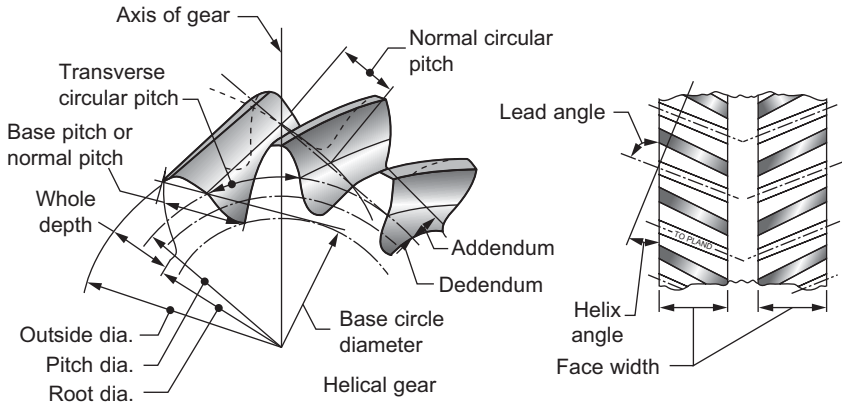


Figure 14-3 Helical gear terminology.

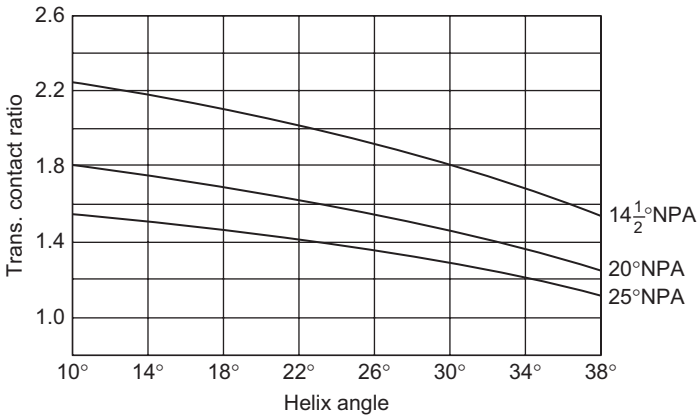


Figure 14-4 Variation of gear tooth geometry with pressure angle and helix angle (courtesy Lufkin Industries, Inc.).

prediction, (5) gear accuracy, (6) bearing types, (7) service factor, (8) gear housing, and (9) lubrication.

Pressure Angle

The decision regarding the pressure angle is one which the designer has to make early in the design stage. Conventionally, pressure angles ranged between 14.5° and 25°. Changes in the pressure angle affect both the contact ratio and the length of line of action. As the pressure angle increases, the contact ratio and the length of line of action decrease as shown in Figures 14-4 and 14-5, respectively. The contact ratio is an indication of the number of teeth in contact. As a general rule, the higher the contact ratio, the lesser the noise the gears generate.

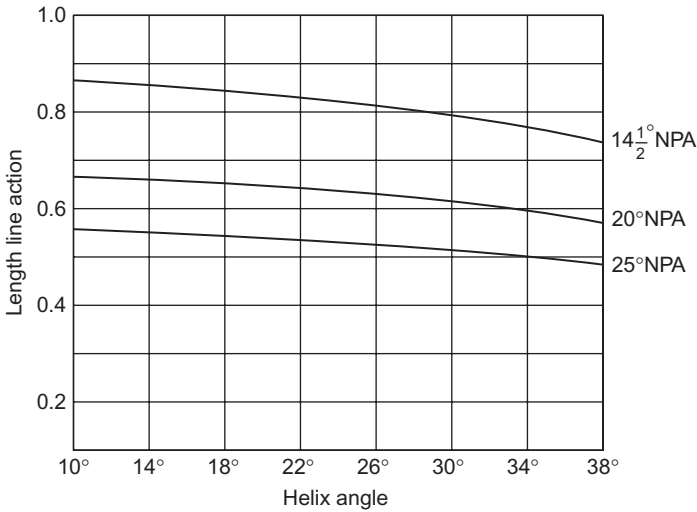


Figure 14-5 Length of line of action change with pressure angle and helix angle (courtesy Lufkin Industries, Inc.).

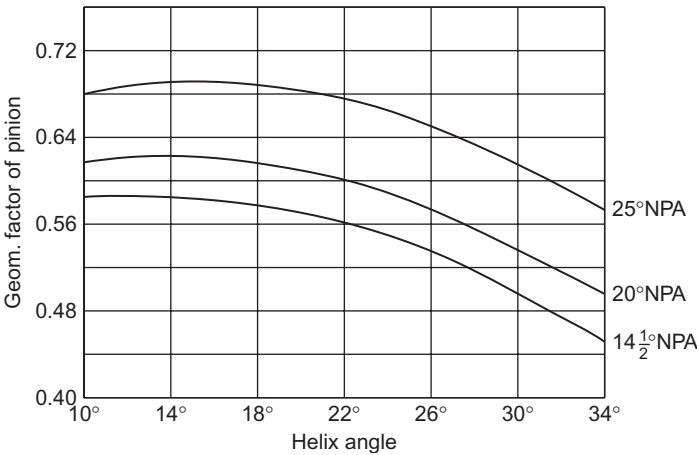


Figure 14-6 Variation of gear tooth geometry with pressure angle and helix angle (courtesy Lufkin Industries, Inc.).

The strength of the tooth is an important factor in the selection of the pressure angle. Figure 14-6 shows the variation of gear tooth geometry and pressure angle. The higher the pressure angle, the higher the tooth strength. The noise that the gears generate decreases with an increase in the contact ratio. Thus, the selection of pressure angle involves many factors. Normal pressure angles in use today are between 17.5° and 22.5°. Higher pressure angles increase the bearing loadings, but this increase is not a determining factor when selecting pressure angles.

Helix Angle

Helix angles vary from 5° to 45° . Single-helical angles fall between 5° and 20° , and double-helical angles fall between 20° and 45° . Helix angles are selected to obtain a minimum overlap ratio and to provide good load sharing. Figure 14-7 shows the effect on overlap ratio with an increasing helix angle. The thrust generated is also a function of the helix angle as shown in Figure 14-8. An increase in the helix angle increases the thrust; thus, this increase is the main reason for the lower helix angles in single-helical gearing.

Both single- and double-helical gearings have advantages and disadvantages. The advantages of single-helical gearing are greater accuracy, less sensitivity to coupling thrust, no apex runout, and less-expensive tooth cutting. The disadvantages of

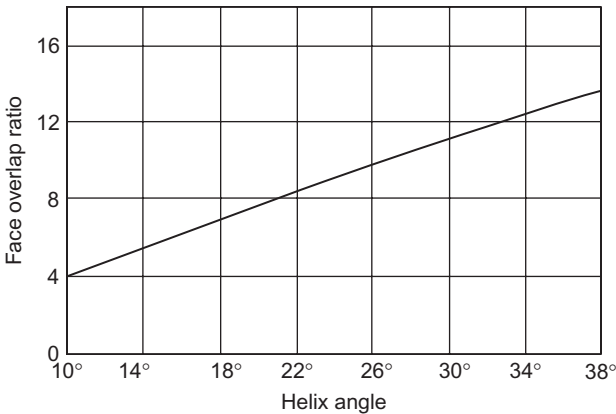


Figure 14-7 Face overlap (contact) ratio variation with helix angle (courtesy Lufkin Industries, Inc.).

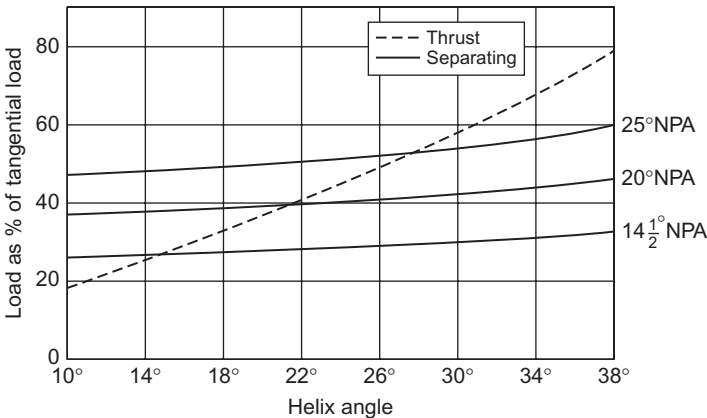


Figure 14-8 Thrust and separating load as a percentage of transmitted tangential load (courtesy Lufkin Industries, Inc.).

single-helical gears are that they require expensive thrust bearings and/or thrust faces and are less-efficient due to heat load on the thrust bearings.

Advantages of double-helical gears are that they are very simple to design and manufacture because of the absence of thrust faces and thrust bearings, very little thrust is produced by gearing, and they are generally more efficient than single-helical gears, which are subject to thrust-bearing losses. Some disadvantages of double-helical gears are as follows: they are sensitive to coupling lockup, they are slightly more expensive to cut teeth for because of setup and tool change, and they usually have a slightly wider gear box to accommodate the gap between helices.

Tooth Hardness

Gears available today have varying hardness, ranging from 225 BHN to 60 Rc. Each gear has its own advantages and disadvantages, so factors determining hardness must be carefully evaluated. Medium-range hardness gears are not too sensitive to operational errors and wear slightly before failing. Also, very hard gears are more susceptible to scuffing due to high-load intensity and sliding velocities. The noise levels on medium-hard gears increase with wear and give warning of gear failure. Heat treatment on medium-hard gears is simple when compared with that on surface-hardened gearing. Hardened gearing is used in turbomachinery applications to provide more power density in a smaller package. When case hardened, gearing must be subjected to a grinding operation to provide the finish. [Figure 14-9](#) shows parts during the carburizing process. Carburizing is a process used to harden low-carbon steels that normally would not respond to quenching and tempering. This is done to provide a tough part with good wear characteristics and high load-carrying capability. Carburizing introduces carbon into a solid ferrous alloy by heating the metal in contact with



Figure 14-9 Parts during the carburizing process (courtesy Lufkin Industries, Inc.).

a carbonaceous material to a temperature above the transformation range and holding at that temperature. The depth of penetration of carbon is dependent on temperature, time at temperature, and the composition of the carburizing agent.

Nitriding is a process that is also used as a case-hardening method for gears. This process provides more load-carrying capability than hardening, but not as much as case carburizing. Other methods of hardening are available, but not in common use for this class of gearing.

Scuffing

For high-speed or high-load intensity, scuffing must be evaluated. One method to predict the probability of scuffing is by using the flash temperature index, calculated from American Gear Manufacturers Association (AGMA) 217. If the index value is less than 275, it is considered to be a low-scuffing risk. Values between 275 and 335 are considered high risks. Scuffing can usually be controlled by using optimum combinations of tooth pitch, lubrication choice, surface finish, and tooth loading.

Figure 14-10 shows the effect of speed and load intensity on the flash temperature index. These curves are general in nature, since scuffing is a function of pressure angle, lubrication, and tooth size.

Gear Accuracy

Turbomachinery gears normally require gear tooth accuracies per ANSI/AGMA ISO 1328-1 of Grade 4 or more. Grinding methods allow close control of tooth lead and profile modification to compensate for rotor and tooth deflections. This provides better load-carrying capabilities by creating more even load distribution. ANSI/AGMA

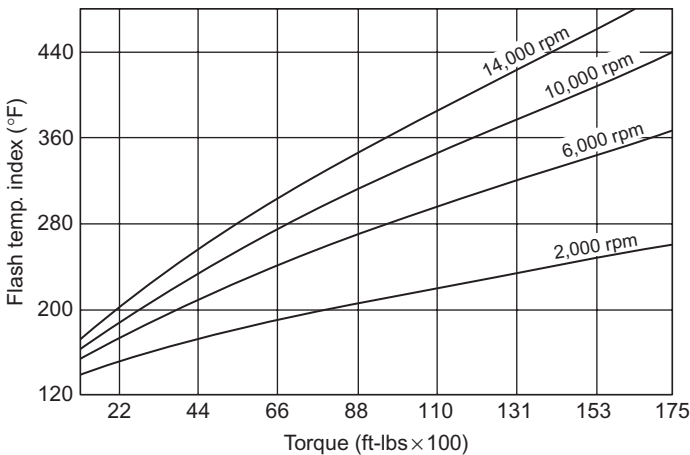


Figure 14-10 Scoring based on flash temperature index related to speed and torque (courtesy Lufkin Industries, Inc.).

ISO 1328-1 provides gear accuracy requirements for individual gears. Care should be taken for high-speed/high-power gears to evaluate the results of mating the measured profiles and leads of the gear and pinion. As a general rule, manufacturers of high-speed gearing monitor the gears and pinions for involute, lead, runout, tooth spacing, and surface finish. Light-load blue transfer checks are conducted to prove the accuracy of the system.

Types of Bearings

Bearings of all types can be used to support gears. Normally, gear drives proposed for turbine-driven applications, whether they are single or double helical, will be supported in hydrodynamic journal bearings.

The most basic type used is plain journal bearings. They have good load carrying capacity, but they can also have oil whirl problems. To prevent oil whirl problems, pressure dam or tilting pad journal bearings are used. Gear motors have imposed operating loads, and do not require the same degree of no-load bearing stability as compressor turbines, which have only the rotor weight applied to the bearings.

The exclusion of rolling-type bearings from drives of this class may be unwarranted. In the low-horsepower ranges, bearing ratings can be easily provided so that race and roller fatigue can be ignored as a source of failure. Drives using rolling element bearings may sometimes provide additional design latitude for the gear manufacturer. The extensive use of rolling element bearings in contemporary light-weight gas turbine designs bears strong testimony for this point.

Thrust bearings vary from the ball bearing to the self-equalizing tilting pad type. The most common type is the babbitt-lined, flat-face thrust bearing. The flat-face bearing is sometimes modified to add tapered lands that double the load-carrying capacity. Tilting pad bearings are becoming more popular because of their high-thrust capacity and misalignment capabilities. Also, the tilting pad thrust bearing is more efficient because of the higher allowable loading and lower rubbing speeds.

Service Factor

When selecting gears, two major areas to consider are the service factor and type of drive to be employed. The service factor is defined as the minimum ratio between calculated capacity and average transmitted load for any component of the system. In general, one of the three criteria will be the controlling influence in gear drives. These are failures due to tooth surface pitting, wear, or physical loss of teeth from breakage. Consequences of the three failure modes differ, particularly in regard to the length of the time involved. Wear can continue for a long period of time without affecting machinery serviceability or reliability. Pitting, if progressive, will eventually destroy the working profile of the teeth, altering their thermal characteristics, and often rendering the drive unsuitable because of high vibration levels long before the teeth are incapable of carrying load. Loss of a portion of a tooth by breakage has immediate

consequences. The balance is immediately and drastically affected and, with major tooth breakage, the gear will be incapable of further operation. Any evaluation of a service factor should determine which of the three modes is involved.

Current practice includes the provision of an additional margin when designing for gear tooth bending. The intent of applying this margin is to eliminate gear tooth breakage as a primary cause of failure, except with severe and unforeseen overloads.

Design against failure by wear under heavy tooth loads will result in the selection of heavy-bodied lubricants, generally 150 SSU or more at supply temperature. Pitting failures are the most difficult to provide a margin for, as increasing gear size or hardness are the only means of improving capacity, and both entail an increase in cost.

The service factor itself is not an overload capacity per se, as it includes either empirical or theoretical estimates of the effect of factors such as length of service life, torque fluctuations, and reliability level required. The service factors as established by the AGMA and the API, along with other organizations, and published in their standards, are intended for application to transmitted load requirements; if substantial overload capacity is planned or allowed (an oversize driver), additional gear rating must be included to provide for operation at those levels. Similarly, torque loads resulting from torsional oscillations or faulty operation are outside the scope of normally applied service factors and must be evaluated and provided for separately. Any torque fluctuations which result in a separation of gear teeth at speed will be most difficult to provide for. Impact loads occur during re-engagement, and very short service life is a frequent result of operation under these conditions, if these loads have not been considered and the gear adequately sized for them.

Gear Housings

Gear housings are made from materials such as cast iron, steel, or aluminum. Before final matching, the gear housing must be stress relieved for dimensional stability. Housing should also be rigid enough to resist misalignment. A sufficient clearance should be provided around the gears to prevent oil choking. To prevent thermal distortion, the design should be able to maintain uniform case temperatures. Gear housings can cause alignment problems from thermal distortion.

Lubrication

The oil furnished to high-speed gears has a dual purpose: tooth and bearing lubrication and cooling. Usually, only 10%–30% of the oil is used for lubrication and 70%–90% is used for cooling.

Turbine oil with rust and oxidation inhibitors is preferred. This oil must be kept clean, cooled, and have the correct viscosity. Synthetic oils are gaining popularity. If their use is anticipated, it is important that the gear manufacturer is informed, so the gear can be designed to be compatible with the synthetic oil.

When 150 SSU at 100°F (38°C) oil is necessary, inlet temperatures should be limited to 110–120°F (43–49°C), if possible, to maintain an acceptable viscosity. Higher

inlet temperatures require more detailed analysis to ensure adequate lubrication to gears. Oil should be supplied in the temperature and pressure range specified by the manufacturer. Up to a pitch line speed of approximately 15,000 ft/min (4,572 m/min), the oil should be sprayed into the outmesh (the side of the mesh where the teeth are disengaging). Spraying the outmesh allows maximum cooling time for the gear blanks and applies the oil at the highest temperature area of the gears. Also, a negative pressure is formed when the teeth come out of the mesh, pulling the oil into the tooth spaces.

At more than approximately 15,000 ft/min (4,572 m/min), 90% of the oil may also be sprayed into the outmesh and 10% into the inmesh. This procedure is a safety precaution to assure that the amount of oil required for lubrication is available at the mesh. When the speed ranges from 25,000 to 40,000 ft/min (7,620–12,192 m/min), oil may be sprayed on the sides and gap area (on double helical) of the gears to minimize thermal distortion.

Manufacturing Processes

Gear manufacturers use several methods for manufacturing good high-speed gearing. The most common are precision hobbing, hobbing and shaving, and hobbing and grinding.

High-speed gearing requires a finishing operation after cutting that involves honing, shaving, or grinding. Typically, gears of this class are hobbled and ground.

Hobbing

Hobbing produces good tooth spacing and accurate lead. [Figure 14-11](#) shows a gear undergoing a hobbing process. It cannot economically achieve a surface finish better than 40 microinches. The cutting tool, called the “hob gear,” is basically a worm gear that has been fluted and has form-relieved teeth as shown in [Figure 14-12](#). These flutes provide the cutting edges and can be sharpened to retain the original tooth profile. As the workpiece meshes with the hob, the teeth are formed by a series of cuts known as the generating process. To cut the helix angle, the rotation of the work is slightly retarded or advanced in relationship to the hob rotation, and the feed is held in a definite relationship with the work and hob. Very accurate gearing can be produced by this process.

Hobbing and Shaving

The shaving process improves surface finish, involute profile, lead, and can be used to crown the teeth. Shaving with inaccurate cutters will reduce hobbled accuracy, and it cannot improve spacing or pitch line runout. The shaving cutter has involute teeth and meshes with the part being shaved, thus improving the finish.



Figure 14-11 The hobbing process for gear manufacture (courtesy Lufkin Industries, Inc.).



Figure 14-12 The hob used in the hobbing process for gear manufacture (courtesy Lufkin Industries, Inc.).

Hobbing and Lapping

Hobbing and lapping was used extensively in the past for this gear class; however, technology has progressed and as the gear materials have become harder in order to carry more load, grinding has replaced lapping as the preferred finishing operation.

Grinding

Grinding produces the best values of lead and involute profile. Figures 14-13 and 14-14 represent a typical grinding machine and a form of grinder, respectively. As a general rule, tooth spacing is good and surface finish can be 24-microinch R_a or more. Current grinding technology includes form grinding, which may be performed with a cubic boron nitride (CBN), aluminum oxide, or ceramic wheel. The advantage of this type of grinder over generating-type grinders is better speed, tooth spacing, and tooth finish, along with highly controllable lead and involute. One drawback of grinding is that the process must be carefully performed and controlled to prevent grinding burns and cracks.



Figure 14-13 Gleason Pfafter form grinder (courtesy Lufkin Industries, Inc.).

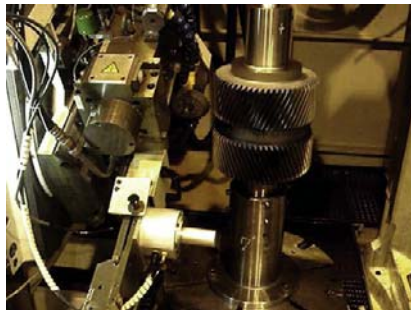


Figure 14-14 A form grinder (courtesy Lufkin Industries, Inc.).

Gear Rating

API 613 “Special Purpose Gear Units for Petroleum, Chemical and Gas Industry Services”, currently Fifth Edition, February 2003, has been used as a worldwide guide for building reliable gears for the turbomachinery market. API 613 provides a simplified gear rating, based on and consistent with ANSI/AGMA 2101-D02. The API 613 rating method is generally more conservative than AGMA methods. A common procedure for comparing and sizing gear is based on the tooth pitting index, the K factor (in U.S. Customary units):

$$\text{Allowable } K = \frac{\text{Material index number}}{\text{Service factor}}$$

The material index is based on hardness. The service factor takes into account the driving and the driven equipment characteristics.

$$K = \frac{126,000 \times P_g}{N_p \times d^2 \times F_w} \times \frac{R + 1}{R}$$

where

P_g = transmitted horsepower

R = gear ratio

N_p = pinion rpm

d = pinion pitch diameter in inches

F_w = net face width of narrowest mating gears, or the sum of the face width of each helix of double helical (inches)

The strength rating based on API 613 is:

$$S = \frac{W_t \times P_n \times SF}{F_w} \times \frac{1.8 \cos \gamma}{J}$$

where,

S = bending stress number

γ = helix angle

J = geometry factor

SF = service factor

P_n = normal diameter pitch

W_t = transmitted tangential load at operating pitch diameter

Gear Noise

Noise from an operating gear set is a function of roundness and concentricity of operating elements (both gearing and shafting), accurate balance, and, in particular, control of tooth spacing errors and uniformity of mesh stiffness to reduce meshing

frequency excitation. Some factors causing gear noise can be attributed to, but not limited to, the following:

- Tooth spacing or involute error
- Contact ratio
- Surface finish
- Wear on tooth flanks or pitting
- Excessive or too little backlash
- Gear, shaft, or housing resonance
- Tooth deflections
- Pitch line runout
- Load intensity on gearing
- Clutches and couplings
- Lube oil pump and piping
- Transmitted noise from driven or driving machinery.

Installation and Initial Operation

The mounting of a gearbox into machinery trains is a precision job and should be done carefully. Gear unit installation is one of the most important factors to be considered for long, trouble-free operation, thus careful testing of the gear box should be undertaken. [Figure 14-15](#) is a gear box under test. No matter how accurately the gear unit is manufactured, it can be destroyed in a few hours of operation when improperly installed. The same care should be taken when installing a gearbox as with any high-speed machinery. The mounting surface should be a flat-level, single-plane surface of finished steel at a height that will permit the shimming necessary to align the gear unit



Figure 14-15 Gas turbine gear drive (courtesy Lufkin Industries, Inc.).

properly to connecting shafts. The shims should be of a size at least equal to the width of the unit foot pad. Then, the gear unit should be placed on the foundation in the approximate required position. Uneven supports can distort the gearbox and adversely affect the gear tooth contact.

Shaft alignment is very important for long gear life. Poor alignment can cause unequal distribution of tooth loads and distortion of the gear elements from overhung moments. A 0.002 inch (0.05 mm) shaft vibration level on the gear unit produced by misalignment is equivalent to a gear pitch line runout of 0.002 inch (0.05 mm).

The gear housing must be properly supported to maintain proper internal gear alignment. When a gear unit is installed, the support pads must be maintained in the same plane as used by the manufacturer during assembly when gear face contact was obtained in the plant. Before startup, gear face contact should be checked using high-spot bluing and rotating or rocking the pinion or lighter element back and forth sharply within the confines of the backlash. Inspection of this blued area should show approximately 90% face contact. If this contact is not obtained, the gear housing can be shimmed under the proper corner until an acceptable face contact is achieved.

Many large, high-hardness or wide-face width gears are manufactured with helix angle modifications to account for torsional and bending deflection. When the helix angle has been modified, 90% face contact will not be obtained under light load. In this case, the gear supplier should furnish data on percent of face contact versus the load to be used as a guide during installation and startup. Also, many gears have a short area of ease-off on each end of the teeth to prevent end loading, and this area usually will not show contact under a light load.

The larger the gear unit, the more important this check becomes, since large housings tend to be more flexible. Also, the use of baseplates furnished by the original equipment manufacturer does not eliminate face contact problems, and these inspection procedures should be carried out.

After the gear checks have been made, the foundation bolts should be uniformly tightened and the alignment should be re-checked. It may be necessary to repeat the shimming and tightening of foundation bolts to obtain final and correct cold alignment.

Alignment of high-speed gear units should always be hot checked and adjustments made as necessary. Temperatures vary so greatly throughout the housing and shafting that it is impossible to calculate a thermal growth accurately and, therefore, an alignment check must be made in the hot condition.

When the alignment is complete, the baseplate or bed should be grouted in as close to the gear housing as possible. Journal bearings are used on the gear shafts and proper oil flow must be maintained. The oil system should therefore be checked thoroughly prior to startup. The gear lube system is normally flushed prior to any operations. Gear mesh spray nozzles should be checked to be sure that dirt was not pumped through the system by observing the sprays or by introducing high-pressure air into the spray nozzle manifold during installation.

When possible, gears should be run-in on initial startup. Speeds and loads should be increased in percentage increments. Lube oil temperature and pressure and bearing temperatures should be observed and adjustments made to the lube system as required.

The number of adjustments made will depend on the complexity of the system. Oil pressure is of primary importance. When an auxiliary pump has been provided, oil should be circulated before the actual start. If not, the pump should be primed, and the journals wetted with oil. Primer holes are sometimes provided as an alternative, and journals can be oiled through the holes provided for bearing temperature detectors.

It is recommended that warning devices be provided to eliminate as much human error as possible, and the set points should be checked carefully. As with any startup, vibrations should be monitored and recorded. The vibration monitoring system should include at least one accelerometer to detect any vibrations generated at gear mesh frequency. The recorded data should be saved to provide baseline vibration data for future reference.

Shutdowns, as well as startups, require care and attention. Shutdown of a unit, which has been operating in a humid atmosphere, can result in considerable condensation and subsequent rusting of the gears, shaft journals, and housing in a very short time. When water contacts the clean steel, it begins to etch the steel immediately. When shutdowns are necessary in such conditions, provisions to prevent condensation must be furnished.

Under normal operating conditions, the oil should be changed every 2,500 operating hours or every 6 months, whichever occurs first. Oil analysis at regular intervals greatly extends the time between oil changes, if utilized correctly. When operating conditions warrant, this period may be extended; conversely, severe operating conditions may make it necessary to change oil at more frequent intervals. Such conditions may occur with the rapid rise and fall in temperature that produces condensation, when operating in moist or dusty atmospheres, or in the presence of chemical fumes. In any case, the lubrication supplier should be consulted when determining a lubricant maintenance program.

Gear Failures

Gears, due to their metal to metal contact, are the most susceptible to problems in many gas turbine trains. Gear teeth can undergo very high pressures that can be further enhanced due to misalignment and loss of lubrication properties of the oil, such as loss of viscosity, due to a mixture of water with the oil or on increased particle count in the oil.

Careful analysis of gear tooth failure evidence may result in valuable information about the mode of failure, which may be used to prevent a recurrence. This analysis may include: visual observation, at the site or in the laboratory; metallurgical examination of microstructure and material properties, physical specimen testing for mechanical properties, testing for chemical properties, load analysis, investigation of installation results, and lubrication analysis, among others.

Figure 14-16 shows a sectional view of a sub-case fatigue that initiated at the case-core interface of a carburized part. This failure was caused by high tooth loading that initiated fatigue failure at a non-metallic inclusion.



Figure 14-16 Damage to gear tooth due to high tooth loads (courtesy Lufkin Industries, Inc.).



Figure 14-17 Plan view of tooth loss due to sub-case fatigue failure.

Figures 14-17 and 14-18 are examples of fatigue failure, where a failure may begin when a crack initiates at an area of structural discontinuity or at a localized area of stress concentration. The crack then propagates as the cycles continue, until the fracture ultimately occurs.

Case crushing, shown in Figure 14-19, can result when the carburized case has insufficient strength to withstand the loading. This may also occur because of insufficient hardness and/or strength in the material core.



Figure 14-18 Root bending fatigue failures of multiple teeth.

Case crushing



Figure 14-19 Case crushing of carburized part due to insufficient case depth and soft core.

The ANSI/AGMA publication, ANSI/AGMA 1010-E95, *Appearance of Gear Teeth-Terminology of Wear and Failure*, provides valuable information on gear tooth failure modes and their identifying features.

Acknowledgement

The author wishes to acknowledge and thank Ms. Lisa Ford, Director of Engineering, Lufkin Industries, for her knowledge, and helpful suggestions in this chapter.

Bibliography

- API 617, *Axial and Centrifugal Compressors and Expander-Compressors for Petroleum, Chemical and Gas Industry Services*, 7th edition, Washington DC, AIP, 2002.
- API 613, *Special Purpose Gear Units for Petroleum, Chemical and Gas Industry Services*, 5th edition, Washington, DC, AIP, 2003.
- Partridge, J.R. High-Speed Gears-Design and Application, *Proceedings of the 6th Turbomachinery Symposium*, Texas A&M Univ., December 1977, pp. 133–142.
- AGMA/ISO 1328-1, *Cylindrical Gears-ISO System of Accuracy-Part 1*, 1999.
- Phinney, J.M. Selection and Application of High-Speed Gear Drives, *Proceedings of the 1st Turbomachinery Symposium*, Texas A&M Univ., October 1972, pp. 62–66.
- AGMA 2101/2001-DO4 *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, 2010.
- ANSI/AGMA 1010-E95, *Appearance of Gear Teeth Terminology of Wear and Failure*, 1995.